



Your Logo

Reproducibility Under Scrutiny: A Critical Assessment of RStudio and Jupyter Notebooks in Dutch Supply Chain Sustainability Innovation Labs

A Graduation Project for the Business Innovation Programme

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Executive Summary

The reproducibility crisis in academic and applied research threatens the credibility of sustainability innovations in supply chain management. This thesis investigates the extent to which open-source computational tools—specifically RStudio and Jupyter Notebooks—can enable reproducible research practices within the Value Chain Hackers (VCH) Lab at Windesheim University of Applied Sciences.

Through a comprehensive case study approach, this research empirically tests the reproducibility of supply chain sustainability interventions, identifies systemic barriers across technical, organizational, and contextual domains, and proposes a structured reproducible research framework aligned with the Enterprise Data Reference Model (EDRM) and FAIR principles.

The findings reveal that while RStudio and Jupyter Notebooks offer powerful capabilities for transparent and repeatable workflows, their effectiveness is severely constrained by institutional factors including inadequate documentation standards, absence of version control practices, ephemeral data storage, and limited methodological training. Analysis of five VCH Lab research tracks (Bakery, CRM, Organizational, Textile, and Cacao) demonstrates that reproducibility failures are systemic rather than individual, stemming from educational models that prioritize creative outputs over process transparency.

This thesis contributes both theoretical insights into reproducibility challenges in business innovation contexts and practical deliverables including a working prototype infrastructure integrating JupyterHub, Quarto, Git/GitHub, and n8n automation. The proposed framework provides Windesheim with an actionable pathway to transform applied research processes into verifiable, cumulative knowledge systems capable of supporting evidence-based sustainability innovations.

Keywords: reproducible research, computational notebooks, supply chain sustainability, Value Chain Hackers, RStudio, Jupyter, FAIR principles, applied research methodology

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To the open-source community maintaining RStudio, Jupyter, Quarto, and the broader ecosystem of reproducible research tools, whose work makes transparent science accessible to applied research contexts.

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This work was made possible by the collaborative spirit of the Value Chain Hackers program and the shared commitment to improving research quality in supply chain sustainability.

List of Abbreviations

Abbreviation	Full Term
API	Application Programming Interface
BI	Business Innovation
CI/CD	Continuous Integration/Continuous Deployment
CRM	Critical Raw Materials
CSV	Comma-Separated Values
DOI	Digital Object Identifier
EDRM	Enterprise Data Reference Model
FAIR	Findable, Accessible, Interoperable, Reusable
Git	Distributed Version Control System
GP1	Graduation Project Plan
GP2	Graduation Project Research Report
HTML	HyperText Markup Language
IDE	Integrated Development Environment
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
PDF	Portable Document Format
R	Statistical Programming Language
SQL	Structured Query Language
UI/UX	User Interface/User Experience
URL	Uniform Resource Locator
VCH	Value Chain Hackers
YAML	YAML Ain't Markup Language

1 Chapter 1: Introduction

1.1 1.1 Background and Context

The credibility of research—both academic and applied—depends fundamentally on its reproducibility. Reproducible research ensures that analyses, findings, and conclusions can be independently verified, validated, and built upon by other researchers (**baker2016?**). When research cannot be reproduced, trust erodes, resources are wasted, and decision-making becomes compromised (**goodman2016?**). This crisis of confidence has become particularly acute in computational research, where even studies claiming to use “open data” frequently fail independent reproduction attempts (**stodden2018?**; **collberg2016?**).

In the domain of supply chain sustainability, reproducibility takes on additional urgency. Organizations, policymakers, and consumers increasingly demand verifiable evidence that sustainability interventions deliver genuine environmental and social impacts (**montabon2016?**). Without reproducible research practices, businesses risk implementing ineffective or misleading practices, regulators lack reliable evidence for policy development, and the broader sustainability movement loses credibility (**ioannidis2005?**).

The Value Chain Hackers (VCH) Lab at Windesheim University of Applied Sciences exemplifies both the opportunities and challenges of applied sustainability research. Established to “revolutionize the supply chain industry through relentless innovation, spirited collaboration, and hands-on practical application,” the VCH Lab brings together students, academic researchers, and industry partners to tackle real-world supply chain challenges. However, preliminary investigation of past VCH Lab projects reveals systematic reproducibility failures: missing documentation, untraceable analytical processes, ephemeral data storage, and absence of version control practices.

These failures are not unique to VCH Lab. They reflect broader institutional patterns in applied research environments where creative problem-solving is prioritized over methodological rigor, where educational models reward outputs rather than transparent processes, and where technical infrastructure for reproducible workflows remains underdeveloped (**freire2012?**).

1.2 1.2 The Reproducibility Crisis: From Science to Business

The reproducibility crisis emerged into public consciousness through high-profile failures in biomedical research, psychology, and computational science (**baker2016?**). A 2016 survey of 1,500 scientists found that more than 70% had failed to reproduce another researcher’s work, and more than 50% had failed to reproduce their own results. While initially framed as a crisis of scientific integrity, reproducibility challenges extend far beyond academic laboratories.

In business innovation and applied research contexts, irreproducible findings carry direct economic and social consequences:

1. **Resource waste:** Organizations invest in interventions based on research that cannot be validated
2. **Regulatory uncertainty:** Policymakers lack reliable evidence for informed decision-making
3. **Stakeholder distrust:** Industry partners, funders, and communities lose confidence in research outputs
4. **Knowledge fragmentation:** Insights cannot accumulate across projects or teams

Supply chain sustainability research faces particular reproducibility challenges due to:

- **Data confidentiality:** Business-sensitive information limits data sharing (**freire2012?**)
- **Contextual complexity:** Supply chain interventions depend on specific organizational, geographic, and temporal contexts
- **Multi-stakeholder dynamics:** Research involves diverse actors with competing interests
- **Methodological pluralism:** Applied research integrates qualitative and quantitative methods in iterative cycles

1.3 1.3 The Promise of Computational Notebooks

Open-source computational notebooks—particularly RStudio and Jupyter Notebooks—have emerged as potential solutions to reproducibility challenges. These tools enable researchers to create “literate programming” documents that interweave narrative text, executable code, data visualizations, and analytical outputs in a single, shareable format (**rule2019?**).

The theoretical advantages are compelling:

- **Transparency:** All analytical steps are visible and documented
- **Executability:** Code can be re-run to verify results
- **Shareability:** Notebooks can be distributed via repositories like GitHub

- **Integration:** Notebooks support multiple programming languages and data formats
- **Accessibility:** Open-source tools reduce barriers to entry

However, the mere availability of these tools does not guarantee reproducible research. As (sandve2013?) observe, “the ability to perform computational research reproducibly is independent from the actual performance of computational research reproducibly.” Tool adoption requires complementary changes in institutional practices, educational training, workflow design, and quality assurance mechanisms.

1.4 1.4 The Value Chain Hackers Lab as Research Context

The VCH Lab provides an ideal context for investigating reproducibility in applied business innovation research. Characteristics include:

Organizational structure: The lab operates at the intersection of academia and industry, with projects commissioned by external partners addressing genuine sustainability challenges in sectors including agriculture, textiles, electronics, and food systems.

Student-led research: Projects are primarily executed by final-year students in 4-6 month cycles, creating natural constraints around knowledge transfer, documentation practices, and methodological consistency.

Innovation focus: The lab’s mission emphasizes creative problem-solving and rapid prototyping, potentially creating tension with slower, more rigorous research documentation practices.

Multi-track design: Parallel research tracks (Bakery, CRM, Organizational, Textile, Cacao) provide opportunities for comparative analysis of reproducibility challenges across different domains.

Existing infrastructure: The lab has access to JupyterHub, Git/GitHub, Quarto publishing tools, and workflow automation platforms (n8n), providing technical foundations for reproducible research.

Preliminary investigation of five completed VCH Lab tracks reveals systematic reproducibility failures:

- **Bakery track:** Single audio recording without transcript, metadata, or analytical documentation
- **CRM track:** Conceptual innovation lacking methodological transparency or version control
- **Organizational track:** Pedagogical model prioritizing creativity without explicit reproducibility requirements

- **Textile track:** Complete data volatility due to dependence on ephemeral cloud tools (Canva, Miro)
- **Cacao track:** Polished outputs lacking documented source selection, analytic synthesis, or validation processes

These patterns suggest that reproducibility failures are systemic rather than individual, requiring institutional-level interventions rather than student-focused remediation.

1.5 1.5 Research Gap and Contribution

While extensive literature addresses reproducibility in academic computational science (stodden2018?; collberg2016?; sandve2013?), limited research examines reproducibility challenges in applied business innovation contexts, particularly in sustainability-focused labs operating at the academia-industry interface.

This thesis addresses three interconnected gaps:

Empirical gap: Most reproducibility studies focus on academic publications rather than applied research conducted in educational innovation labs. This research provides systematic documentation of reproducibility barriers across multiple real-world supply chain sustainability projects.

Methodological gap: Existing reproducibility frameworks (FAIR principles, Open Science guidelines) are designed primarily for academic research. This thesis adapts and extends these frameworks for business innovation contexts where data confidentiality, contextual specificity, and stakeholder sensitivities constrain full transparency.

Practical gap: Limited guidance exists for institutions seeking to build reproducible research capacity in applied settings. This thesis delivers a working prototype infrastructure and implementation roadmap specifically designed for the constraints and opportunities of student-led, industry-partnered innovation research.

The research contributes:

1. **Empirical evidence** of reproducibility challenges in applied supply chain sustainability research
2. **Theoretical insights** into the interplay of technical, organizational, and contextual barriers
3. **Practical deliverables** including a functional reproducible research infrastructure and implementation guidelines
4. **Institutional recommendations** for embedding reproducibility into educational and research workflows

1.6 1.6 Research Objectives and Questions

1.6.1 Primary Objective

To investigate, implement, and document a reproducible research framework for the Value Chain Hackers Lab that enables transparent, verifiable, and cumulative knowledge development in supply chain sustainability innovation.

1.6.2 Research Questions

RQ1: To what extent can RStudio and Jupyter Notebooks enable reproducible research in real-world supply chain sustainability projects?

This question examines the technical capabilities and limitations of computational notebooks when applied to the messy realities of applied innovation research. It considers both the theoretical affordances of these tools and their practical performance given constraints of data confidentiality, stakeholder complexity, and iterative research processes.

RQ2: What are the main technical, organizational, and contextual obstacles to achieving reproducibility in this domain?

This question identifies and categorizes barriers systematically, distinguishing between: - Technical barriers (tool limitations, infrastructure gaps) - Organizational barriers (institutional policies, workflow standards, training deficiencies) - Contextual barriers (data sensitivity, stakeholder dynamics, business constraints)

RQ3: How do issues such as data confidentiality, tool limitations, and context specificity affect the transferability and credibility of reproducible research findings?

This question examines second-order effects: even when research is technically reproducible, how do business realities constrain the types of transparency and verification that are feasible or desirable? It considers the trade-offs inherent in applied research contexts.

1.7 1.7 Methodology Overview

This research employs a case study methodology with design science components:

Case study analysis: Systematic retrospective analysis of five completed VCH Lab tracks to identify reproducibility failures, categorize barriers, and extract lessons learned.

Design science: Development and prototyping of a reproducible research infrastructure integrating existing VCH Lab tools (JupyterHub, Quarto, Git/GitHub, n8n) following Enterprise Data Reference Model (EDRM) principles.

Reproducibility audit: Empirical testing of the infrastructure by attempting to reproduce analysis using provided materials and documentation.

Stakeholder consultation: Interviews with VCH Lab participants, supervisors, IT staff, and industry partners to understand practical barriers and requirements.

Iterative refinement: Multiple cycles of prototype development, testing, feedback, and revision.

Detailed methodology is presented in Chapter 3.

1.8 1.8 Scope and Limitations

1.8.1 Scope

This research focuses specifically on:

- Computational reproducibility (ability to regenerate results from code and data)
- Supply chain sustainability projects within the VCH Lab context
- Open-source tools accessible to educational institutions
- Processes that can be implemented at institutional scale

1.8.2 Limitations

This research does not address:

- Full scientific replication (conducting new studies to validate findings)
- Proprietary commercial tools requiring licensing
- Large-scale enterprise data infrastructure beyond educational lab scope
- Reproducibility in purely qualitative research not involving computational analysis

1.8.3 Delimitations

The research deliberately limits focus to the VCH Lab to enable deep contextual understanding and practical implementation. Findings are expected to transfer to similar applied research contexts (innovation labs, practice-oriented research programs, industry-academic partnerships) but are not claimed to generalize to all research domains.

1.9 1.9 Significance and Impact

1.9.1 Academic Significance

This research contributes to emerging literature on reproducible research in applied contexts, extending reproducibility frameworks beyond traditional academic science into business innovation and sustainability domains. It provides empirical evidence of the specific challenges that arise when reproducibility principles encounter organizational realities, data sensitivities, and stakeholder dynamics.

1.9.2 Practical Significance

The deliverables from this research—including the reproducible research infrastructure prototype, implementation guidelines, and diagnostic framework—provide actionable resources for institutions seeking to strengthen research quality. The approach is designed for scalability and adaptation across different applied research contexts.

1.9.3 Social Significance

By strengthening the verifiability and credibility of supply chain sustainability research, this work supports more evidence-based decision-making by businesses, policymakers, and civil society organizations. Reproducible research practices reduce the risk of implementing ineffective or misleading sustainability interventions, thereby contributing to genuine environmental and social impact.

1.10 1.10 Thesis Structure

The remainder of this thesis is organized as follows:

Chapter 2: Literature Review synthesizes existing research on reproducibility challenges, computational notebooks, supply chain sustainability research, and institutional barriers to open science practices.

Chapter 3: Methodology details the case study design, infrastructure development approach, data collection methods, and analytical procedures.

Chapter 4: Case Study Analysis presents systematic analysis of reproducibility failures across five VCH Lab tracks, identifying patterns and categorizing barriers.

Chapter 5: Infrastructure Design and Implementation documents the development of the reproducible research framework, including architecture decisions, tool integration, workflow design, and prototype testing.

Chapter 6: Findings and Discussion synthesizes insights across case studies and implementation work, addressing each research question and discussing implications for theory and practice.

Chapter 7: Conclusions and Recommendations summarizes key findings, acknowledges limitations, and provides concrete recommendations for VCH Lab, Windesheim University, and the broader applied research community.

Chapter 8: Reflection offers personal reflection on the learning process, meta-skills development, and future research directions.

2 Chapter 2: Literature Review

2.1 2.1 Introduction to the Literature Review

This literature review synthesizes scholarly work across four interconnected domains relevant to reproducible research in supply chain sustainability innovation contexts:

1. **The reproducibility crisis:** Origins, manifestations, and proposed solutions
2. **Computational notebooks and open science tools:** Technical capabilities and adoption barriers
3. **Supply chain sustainability research:** Methodological challenges and verification needs
4. **Institutional factors in research quality:** Organizational, cultural, and educational dimensions

The review identifies theoretical foundations, empirical evidence, and conceptual gaps that inform this thesis’s research questions and methodological approach.

2.2 2.2 The Reproducibility Crisis in Science and Applied Research

2.2.1 2.2.1 Defining Reproducibility

(goodman2016?) distinguish three forms of reproducibility:

- **Methods reproducibility:** Providing sufficient detail to enable independent researchers to repeat procedures
- **Results reproducibility:** Obtaining consistent results using the same methods on independent data
- **Inferential reproducibility:** Drawing consistent conclusions from either the same or different data

This thesis focuses primarily on methods reproducibility in computational contexts: can an independent analyst regenerate reported results using the provided code, data, and documentation?

2.2.2 2.2.2 The Emergence of the Reproducibility Crisis

The reproducibility crisis gained widespread attention through (baker2016?)’s survey of 1,500 scientists, revealing that 52% agreed there is a “significant reproducibility crisis.” Contributing factors identified in the literature include:

Publication bias: Pressure to publish “positive” results creates incentives to selectively report findings (ioannidis2005?)

Methodological complexity: Increasing sophistication of computational analysis creates opportunities for errors and ambiguities (collberg2016?)

Inadequate training: Researchers receive limited education in reproducible workflow practices (sandve2013?)

Tool proliferation: Rapid evolution of software tools creates dependency management challenges (stodden2018?)

Cultural norms: Academic reward structures prioritize novelty over verification (nosek2015?)

2.2.3 2.2.3 Reproducibility in Computational Research

(stodden2018?) conducted an empirical analysis of computational reproducibility in published research, finding that even when code and data are “available,” substantial researcher time and expertise are required to reproduce results. Common barriers include:

- Missing dependencies or software versions
- Inadequate documentation of computational environments
- Proprietary software requirements
- Unclear file organization
- Undocumented data preprocessing steps

(collberg2016?) attempted to reproduce 613 computational systems research papers, successfully building systems for only 32.1% of papers. Barriers included code unavailability, missing build instructions, and undocumented dependencies.

2.2.4 2.2.4 Reproducibility Beyond Traditional Science

While most reproducibility literature focuses on academic research, (freire2012?) examine reproducibility challenges in business and government contexts, identifying additional constraints:

- **Data confidentiality:** Business-sensitive information cannot be publicly shared

- **Stakeholder accountability:** Multiple parties have competing interests in research transparency
- **Regulatory compliance:** Legal and ethical frameworks constrain data handling
- **Cost structures:** Applied research often lacks funding for extensive documentation

These constraints require adapted reproducibility frameworks that balance transparency ideals with practical realities.

2.3 2.3 Solutions and Frameworks for Reproducible Research

2.3.1 2.3.1 The FAIR Principles

(wilkinson2016?) proposed the FAIR principles for scientific data management:

- **Findable:** Persistent identifiers, rich metadata
- **Accessible:** Retrievable via standard protocols
- **Interoperable:** Standards-compliant formats and vocabularies
- **Reusable:** Clear licensing, provenance, and documentation

Originally designed for data repositories, FAIR principles increasingly guide broader reproducible research infrastructure development.

2.3.2 2.3.2 Ten Simple Rules for Reproducible Computational Research

(sandve2013?) distilled practical guidance:

1. For every result, keep track of how it was produced
2. Avoid manual data manipulation
3. Archive exact versions of all external programs
4. Version control all custom scripts
5. Record all intermediate results
6. For analyses including randomness, note underlying seeds
7. Store raw data behind plots
8. Generate hierarchical analysis output
9. Connect textual statements to underlying results
10. Provide public access to scripts, runs, and results

These rules provide actionable steps for individual researchers but assume supportive institutional infrastructure.

2.3.3 Computational Notebooks as Reproducibility Tools

(rule2019?) analyze Jupyter Notebooks as reproducibility infrastructure, proposing “ten simple rules for writing and sharing computational analyses in Jupyter Notebooks”:

1. Tell a story for an audience
2. Document the process, not just the results
3. Use cell divisions to organize steps
4. Modularize code
5. Record dependencies
6. Use version control
7. Build a pipeline
8. Share and explain your data
9. Design your notebooks to be read, run, and explored
10. Advocate for open research

(kluyver2016?) describe Jupyter’s architecture and the “computational narrative” paradigm, emphasizing the integration of explanation, code, and outputs.

2.3.4 Infrastructure and Workflow Tools

Reproducible research requires integrated toolchains (stodden2018?):

Version control systems (Git, GitHub) track changes to code and documentation, enable collaboration, and provide history

Environment management (Docker, conda, renv) ensure computational environments can be reconstructed

Literate programming tools (Jupyter, R Markdown, Quarto) integrate code, narrative, and outputs

Workflow automation (Make, Snakemake, n8n) reduce manual steps and document dependencies

Data versioning (DVC, Git-LFS) track changes to datasets

Publication platforms (GitHub Pages, Binder, Quarto Pub) share interactive computational artifacts

2.4 2.4 Supply Chain Sustainability Research: Challenges and Opportunities

2.4.1 2.4.1 The Imperative for Evidence-Based Sustainability

(montabon2016?) argue that “making sustainability sustainable” requires rigorous validation of intervention effectiveness. They identify common failures:

- Confusing activity with impact
- Relying on anecdotal evidence
- Failing to measure long-term outcomes
- Neglecting unintended consequences

Reproducible research practices address these failures by enabling independent verification of sustainability claims.

2.4.2 2.4.2 Methodological Pluralism in Supply Chain Research

Supply chain sustainability research typically integrates:

- **Quantitative analysis:** Life cycle assessment, carbon footprinting, cost-benefit analysis
- **Qualitative investigation:** Stakeholder interviews, case studies, ethnography
- **Participatory methods:** Co-design, action research, multi-stakeholder processes

This methodological diversity complicates reproducibility, as different methods require different verification approaches (montabon2016?).

2.4.3 2.4.3 Data Challenges in Supply Chain Research

Supply chain data are characterized by:

- **Distributed ownership:** Data held by multiple organizations
- **Commercial sensitivity:** Competitive concerns limit sharing
- **Heterogeneous formats:** Lack of standardization across systems
- **Dynamic contexts:** Supply chains evolve rapidly

These characteristics require reproducibility frameworks that accommodate partial transparency and contextual specificity (freire2012?).

2.5 2.5 Institutional and Organizational Barriers to Reproducible Research

2.5.1 2.5.1 Academic Culture and Incentive Structures

([nosek2015?](#)) analyze how academic reward systems prioritize publication quantity and novelty over verification and robustness. Career progression depends on “positive” results, creating disincentives for time-intensive documentation and validation work.

([munaf02017?](#)) call for a “manifesto for reproducible science,” arguing that cultural change requires:

- Protecting against cognitive biases
- Improving methodological training
- Implementing independent verification
- Encouraging collaboration and data sharing
- Aligning incentives with open practices

2.5.2 2.5.2 Educational Gaps

([baumer2014?](#)) survey data science education, finding that reproducible workflow practices receive limited attention in curricula. Students learn statistical methods and programming but not version control, documentation standards, or workflow automation.

([wilson2014?](#)) document “best practices for scientific computing” but acknowledge that most researchers are self-taught programmers lacking formal software engineering training.

2.5.3 2.5.3 Institutional Infrastructure Deficiencies

Even when individual researchers are motivated to adopt reproducible practices, institutional barriers persist ([stodden2018?](#)):

- Lack of centralized data storage and compute infrastructure
- Absence of technical support for reproducibility tools
- Inadequate policies around data management and sharing
- Limited training resources
- Competing IT priorities

2.5.4 The “Valley of Death” Between Tools and Adoption

(freire2012?) identify a persistent gap between the availability of reproducibility tools and their actual adoption. Barriers include:

- **Learning curves:** Tools require substantial upfront investment
- **Workflow disruption:** Adopting new practices interrupts established routines
- **Perceived low return on investment:** Benefits accrue to others (readers, future researchers) rather than original authors
- **Lack of community norms:** Without field-wide standards, individual adoption is costly

2.6 Applied Research and Practice-Oriented Science

2.6.1 Characteristics of Applied Research Contexts

Applied research differs from basic science in several dimensions relevant to reproducibility:

- **Problem-driven:** Research questions emerge from practical challenges rather than theoretical gaps
- **Stakeholder-embedded:** Multiple parties have interests in methods and findings
- **Iterative and adaptive:** Methods evolve as understanding develops
- **Time-constrained:** Deadlines may limit documentation thoroughness
- **Multi-disciplinary:** Teams integrate diverse expertise and methodological traditions

2.6.2 Universities of Applied Sciences Context

Dutch universities of applied sciences (hogescholen) emphasize “practice-oriented research” bridging academic knowledge and professional practice. Characteristics include:

- **Student involvement:** Research often conducted by students as capstone projects
- **Industry partnerships:** Research commissioned by external organizations
- **Knowledge transfer focus:** Emphasis on actionable insights over theoretical contribution
- **Short-term cycles:** Projects typically span 4-6 months

This context creates both opportunities (real-world relevance) and challenges (limited resources, high turnover) for reproducible research.

2.6.3 2.6.3 Innovation Labs as Research Infrastructure

Innovation labs provide dedicated spaces for collaborative, experimental research. (chesbrough2003?) describe how “open innovation” models depend on knowledge sharing across organizational boundaries, requiring transparent processes and documented artifacts.

The VCH Lab exemplifies this model, bringing together students, faculty, and industry partners. However, (freire2012?) observe that innovation labs often prioritize speed and creativity over documentation and verification, creating reproducibility tensions.

2.7 2.7 Enterprise Data Reference Model and Data Governance

2.7.1 2.7.1 The EDRM Framework

The Enterprise Data Reference Model provides architectural guidance for managing data assets consistently across an organization. Core principles include:

- **Standard definitions:** Common data vocabularies and ontologies
- **Clear ownership:** Assigned responsibility for data quality and governance
- **Lifecycle management:** Defined processes for data creation, use, archival, and deletion
- **Quality assurance:** Validation and verification mechanisms
- **Access control:** Security and privacy protections

While EDRM is typically applied in enterprise IT contexts, its principles translate to reproducible research infrastructure design.

2.7.2 2.7.2 ISO 27001 and Data Security

ISO 27001 provides an information security management framework addressing:

- Confidentiality controls
- Integrity assurance
- Availability guarantees
- Access management
- Audit logging

For reproducible research in business contexts, security frameworks enable appropriate transparency without compromising sensitive data (freire2012?).

2.8 2.8 Gaps in the Literature

Synthesis of the literature reveals several gaps this thesis addresses:

Empirical gap: Most reproducibility studies examine academic publications rather than applied research in innovation labs. Limited systematic documentation exists of reproducibility barriers in practice-oriented, student-led, industry-partnered research.

Methodological gap: Existing reproducibility frameworks (FAIR, Open Science) are designed primarily for academic contexts with public data sharing assumptions. Adaptation is needed for business innovation contexts with confidentiality constraints.

Practical gap: While many studies identify reproducibility barriers, fewer provide detailed implementation guidance for building reproducible research capacity at institutional scale in resource-constrained settings.

Contextual gap: Limited research examines reproducibility in supply chain sustainability domains where methodological pluralism, stakeholder complexity, and contextual specificity create unique challenges.

2.9 2.9 Theoretical Framework for This Research

Drawing on the literature, this thesis adopts a multi-level framework for understanding reproducibility:

Technical level: Tool capabilities, infrastructure, and computational environments

Methodological level: Research design, documentation practices, and verification procedures

Organizational level: Institutional policies, training programs, and resource allocation

Cultural level: Norms, incentives, and values shaping researcher behavior

Reproducibility emerges from alignment across these levels. Failures occur when technical capabilities exist but organizational and cultural supports are absent.

This framework informs the case study analysis (Chapter 4), infrastructure design (Chapter 5), and recommendations (Chapter 7).

2.10 2.10 Chapter Summary

This literature review established that:

1. Reproducibility crisis is well-documented in academic science and extends to applied research contexts
2. Computational notebooks offer promising but not sufficient solutions
3. Reproducibility requires integrated technical infrastructure, methodological standards, organizational policies, and cultural norms
4. Supply chain sustainability research faces unique challenges balancing transparency with confidentiality
5. Applied research contexts like innovation labs have characteristics requiring adapted reproducibility frameworks
6. Significant gaps exist in empirical documentation and practical implementation guidance for reproducible research in business innovation settings

The following chapters build on this foundation, using case study analysis and design science methods to investigate reproducibility challenges in the VCH Lab context and develop practical solutions.

3 Chapter 3: Methodology

3.1 3.1 Research Design Overview

This research employs a **case study methodology with design science components**, combining retrospective analysis of existing VCH Lab projects with prospective development and testing of a reproducible research infrastructure.

3.1.1 3.1.1 Rationale for Case Study Approach

(yin2018?) defines case study as “an empirical method that investigates a contemporary phenomenon in depth and within its real-world context.” Case study methodology is appropriate for this research because:

1. **Contextual complexity:** Reproducibility challenges are deeply embedded in institutional, organizational, and social contexts
2. **“How” and “why” questions:** The research investigates how reproducibility failures occur and why certain barriers persist
3. **Limited control:** The researcher observes and analyzes existing practices rather than manipulating experimental variables
4. **Multiple evidence sources:** Case studies enable triangulation across documentation, interviews, and artifacts

3.1.2 3.1.2 Rationale for Design Science Components

(hevner2004?) describe design science as “fundamentally a problem-solving paradigm” that “creates and evaluates IT artifacts intended to solve identified organizational problems.” Design science complements case study by:

1. **Producing actionable artifacts:** Creating infrastructure rather than only documenting problems
2. **Iterative refinement:** Enabling cycles of prototyping, testing, and improvement
3. **Demonstration of feasibility:** Proving that proposed solutions actually work in context
4. **Knowledge contribution:** Generating both prescriptive design principles and empirical insights

The combination enables both **diagnosis** (what barriers exist?) and **solution development** (how can they be addressed?).

3.2 3.2 Research Phases and Timeline

The research proceeded through five phases aligned with the graduation project timeline:

3.2.1 Phase 1: Research and Planning (Weeks 1-2)

Activities: - Literature review on reproducibility, computational notebooks, and supply chain sustainability research - Initial analysis of VCH Lab organizational structure and past projects - Stakeholder identification and initial consultations - Development of case study protocols and interview guides - Ethical considerations and confidentiality agreements

Outputs: - Literature synthesis - Research protocol - Case study selection criteria - Interview guide - Timeline and resource plan

3.2.2 Phase 2: Case Study Data Collection (Weeks 3-5)

Activities: - Systematic review of documentation from five VCH Lab tracks (Bakery, CRM, Organizational, Textile, Cacao) - Interviews with VCH Lab coordinator, participating students (where possible), and faculty supervisors - Analysis of existing VCH Lab technical infrastructure (JupyterHub, Git, file storage) - Review of pedagogical materials and assessment criteria

Outputs: - Case study database - Interview transcripts - Documentation audit findings - Infrastructure assessment report

3.2.3 Phase 3: Design and Prototype Development (Weeks 6-10)

Activities: - Application of EDRM principles to VCH Lab workflow design - Architecture design for reproducible research infrastructure - Tool integration (JupyterHub, Quarto, Git/GitHub, n8n) - Development of templates, guidelines, and documentation standards - Iterative prototyping with feedback from VCH Lab coordinator

Outputs: - System architecture documentation - Working prototype infrastructure - Template library - Initial user documentation

3.2.4 Phase 4: Validation and Iteration (Weeks 11-13)

Activities: - Reproducibility audit: independent researcher attempts to reproduce sample analysis - Usability testing with VCH Lab students - Technical debugging and refinement - Documentation improvement based on user feedback - Performance and reliability testing

Outputs: - Reproducibility audit report - Refined infrastructure - Validated templates and workflows - Enhanced documentation

3.2.5 Phase 5: Analysis, Writing, and Finalization (Weeks 14-16)

Activities: - Cross-case synthesis and pattern identification - Analysis of infrastructure implementation learnings - Thesis writing - Creation of final deliverables (repository, Quarto documentation, presentation) - Preparation for oral defense

Outputs: - Complete thesis document - Public GitHub repository - Quarto-published documentation website - Presentation materials - Reflective report

3.3 Case Study Design

3.3.1 Unit of Analysis

The primary unit of analysis is the **individual VCH Lab research track**. Five tracks were selected to provide variation across supply chain domains (food, materials, textiles, organizational), research maturity (completed projects), and methodological approaches (quantitative, qualitative, mixed).

3.3.2 Case Selection Criteria

Cases were selected based on:

1. **Recency:** Completed within past 2 years to enable student contact where possible
2. **Documentation availability:** Sufficient artifacts exist for analysis
3. **Domain diversity:** Cases span different supply chain sectors
4. **Accessibility:** VCH Lab coordinator could provide access to materials
5. **Representativeness:** Cases reflect typical VCH Lab project characteristics

Selected cases:

3 Chapter 3: Methodology

Case	Domain	Duration	Team Size	Primary Methods
Bakery	Food systems	4 months	3 students	Qualitative interviews
CRM	Raw materials	5 months	4 students	Mixed methods
Orga	Meta-organizational	Ongoing	Faculty/students	Pedagogical design
Textile	Apparel	4 months	3 students	Visual/creative methods
Cacao	Agriculture	6 months	4 students	Secondary research synthesis

3.3.3 3.3.3 Data Sources

For each case, data were collected from:

Project documentation: - Proposals, reports, presentations - Data files, code, notebooks - Meeting notes, communication logs - Deliverables submitted to industry partners

Archival materials: - Git repositories (where available) - Shared drives and cloud storage - Email threads - Assessment rubrics and feedback

Interviews: - Semi-structured interviews with VCH Lab coordinator (90 minutes) - Email correspondence with available student alumni (3 responses) - Faculty supervisor consultation (45 minutes)

Direct observation: - Current VCH Lab infrastructure examination - Tool testing (JupyterHub, Git instance) - Pedagogical material review

3.4 3.4 Case Study Data Collection and Analysis

3.4.1 3.4.1 Documentation Audit Protocol

For each case, a systematic documentation audit assessed:

Methodological transparency: - Are research questions clearly stated? - Are data collection methods documented? - Are analytical procedures described? - Are assumptions and limitations acknowledged?

Data integrity: - Are raw data preserved? - Are data transformations documented? - Are data file formats sustainable? - Is data provenance traceable?

Verifiability: - Could an independent researcher understand what was done? - Are computational environments documented? - Is code available and executable? - Are dependencies specified?

Archival sustainability: - Are materials stored in accessible formats? - Are persistent identifiers used? - Is there version control? - Are materials at risk of loss (e.g., student account expiration)?

Each dimension was scored on a 4-point scale: - 0 = Absent or completely inadequate - 1 = Minimal, significant gaps - 2 = Partial, some gaps - 3 = Complete, best practices demonstrated

3.4.2 Interview Protocols

Semi-structured interviews with the VCH Lab coordinator explored:

Current practices: - How are projects initiated and scoped? - What guidance do students receive on research methods? - What tools are students expected to use? - How is quality assessed?

Perceived barriers: - What challenges do students face in documentation? - What technical limitations exist? - What competing priorities affect research quality?

Improvement opportunities: - What changes would be most valuable? - What resources are available for infrastructure development? - What institutional constraints must be navigated?

Interviews were recorded (with consent), transcribed, and coded using thematic analysis.

3.4.3 Cross-Case Analysis

Following individual case analysis, cross-case synthesis identified:

Patterns: Common barriers appearing across multiple cases

Anomalies: Cases where reproducibility was better or worse than typical

Typology: Categorization of barrier types (technical, organizational, contextual)

Root causes: Systemic factors underlying observed failures

Leverage points: Where interventions would have greatest impact

Analysis was iterative, moving between case-specific observations and abstracted patterns.

3.5 3.5 Design Science Methodology for Infrastructure Development

3.5.1 3.5.1 Design Science Framework

Following (Hevner2004?), the infrastructure development applied a rigorous design science approach:

Problem identification and motivation: Documented through case study analysis (Phase 2)

Objectives of a solution: - Enable computational reproducibility for VCH Lab projects - Integrate with existing tools and workflows - Require minimal technical expertise from students - Scale across multiple simultaneous projects - Preserve confidentiality where required

Design and development: Iterative prototyping following EDRM principles and FAIR guidelines

Demonstration: Working prototype tested with realistic VCH Lab scenarios

Evaluation: Reproducibility audit by independent researcher

Communication: Quarto-published documentation, GitHub repository, thesis

3.5.2 3.5.2 EDRM Application to VCH Lab Context

The Enterprise Data Reference Model guided infrastructure design across six domains:

Data architecture: - Standard schemas for metadata - Common vocabularies for VCH domains - File organization conventions

Data governance: - Roles and responsibilities for data quality - Approval workflows for data sharing - Retention and archival policies

Data quality: - Validation rules - Documentation standards - Version control requirements

Data operations: - Backup and recovery procedures - Access control mechanisms - Monitoring and logging

Data security: - Confidentiality classifications - Encryption standards - Ethical compliance checks

Data platforms: - JupyterHub for computation - Git/GitHub for version control - Quarto for publication - n8n for automation

3.5.3 3.5.3 Tool Selection and Integration

Tools were selected based on:

Existing VCH Lab access: Preference for already available platforms

Open-source: No licensing costs or proprietary lock-in

Education-friendly: Suitable for student skill levels

Interoperability: Compatible with each other

Community support: Active development and documentation

Selected tools and their roles:

Tool	Purpose	Rationale
JupyterHub	Computational environment	Already deployed, supports Python and R, enables collaborative work
Quarto	Literate programming & publication	Successor to R Markdown, supports multiple languages, publication-ready outputs
Git/GitHub	Version control	Industry standard, free for education, integrates with many tools
n8n	Workflow automation	Open-source, visual programming, extensible
Docker	Environment reproducibility	Containerization ensures computational environment consistency

3.5.4 3.5.4 Prototype Development Process

Infrastructure development proceeded iteratively:

Sprint 1: Basic integration - Set up Git repository structure - Configure Quarto publication pipeline - Document minimal viable workflow

Sprint 2: Template development - Create project templates - Develop metadata standards - Build example notebooks

Sprint 3: Automation - Implement n8n workflows for repetitive tasks - Automate environment documentation - Configure continuous integration checks

Sprint 4: Refinement - User testing with VCH Lab students - Documentation improvement - Bug fixes and usability enhancements

Each sprint involved design, implementation, testing, and reflection phases.

3.6 3.6 Reproducibility Audit Methodology

To validate the infrastructure, a reproducibility audit tested whether an independent researcher could reproduce analysis following provided documentation.

3.6.1 3.6.1 Audit Protocol

Step 1: Environment setup - Follow documentation to configure computational environment - Install dependencies - Verify access to data and code

Step 2: Execution - Run provided notebooks/scripts - Compare generated outputs to documented results - Document any errors or discrepancies

Step 3: Verification - Check that numerical results match (within floating-point tolerance) - Verify that visualizations are substantially equivalent - Confirm that conclusions are supported by reproduced outputs

Step 4: Documentation assessment - Evaluate clarity and completeness of instructions - Identify missing information or ambiguities - Suggest improvements

3.6.2 3.6.2 Auditor Selection

The reproducibility auditor was selected based on: - Competent in Python or R programming - Familiar with Jupyter/RStudio but not with this specific project - Available to invest 4-6 hours in reproduction attempt - Willing to provide detailed feedback

3.6.3 3.6.3 Success Criteria

Reproducibility was considered successful if: - Auditor could set up environment without external assistance - Code executed without errors - Numerical outputs matched within 1% tolerance - Qualitative outputs (visualizations, tables) were substantially equivalent - Auditor confirmed they understood the analytical process

3.7 3.7 Ethical Considerations

3.7.1 3.7.1 Research Ethics Approval

This research involved human subjects (interviews) and analysis of student work. The following ethical safeguards were implemented:

Informed consent: - Interview participants received information sheets explaining research purpose, data use, and confidentiality protections - Written consent obtained before interviews - Participants informed of right to withdraw

Confidentiality: - Student names removed from case study documentation - Industry partner names anonymized where sensitive - Interview transcripts stored securely - Data sharing limited to aggregate findings

Institutional approval: - Research plan reviewed by Inholland thesis supervisor - VCH Lab coordinator provided institutional consent for case access

3.7.2 3.7.2 Data Management

Research data were managed following FAIR principles:

Findable: - Project repository with DOI (via Zenodo) - Descriptive metadata for all datasets

Accessible: - Public repository on GitHub - Documentation on persistent platform (Quarto Pub)

Interoperable: - Standard file formats (CSV, JSON, Markdown) - Documented data structures

Reusable: - CC-BY license for documentation - MIT license for code - Clear provenance and version information

Sensitive data (interview transcripts, confidential project materials) were stored separately with access restricted to researcher and supervisors.

3.7.3 3.7.3 Conflicts of Interest

The researcher was embedded in the VCH Lab context, creating potential conflicts:

Mitigation strategies: - Explicitly sought disconfirming evidence - Triangulated findings across multiple data sources - Engaged independent auditor for reproducibility testing - Acknowledged positionality and potential biases in reflective writing

3.8 3.8 Quality and Rigor

3.8.1 3.8.1 Validity Strategies

Construct validity (are constructs measured appropriately?): - Multiple data sources for each case (triangulation) - Detailed case study protocol - Explicit operationalization of reproducibility dimensions

Internal validity (are causal inferences justified?): - Pattern matching across cases - Rival explanations considered - Transparent chain of evidence

External validity (are findings generalizable?): - Purposive sampling for variation - Thick description enabling transferability assessment - Explicit scope conditions

Reliability (is study replicable?): - Detailed methodology documentation - Case study database preserved - Audit trail of analytical decisions

3.8.2 3.8.2 Design Science Evaluation Criteria

Infrastructure quality was assessed using (Hevner2004?)’s criteria:

Efficacy: Does the infrastructure enable reproducibility? *Measurement:* Reproducibility audit success rate

Efficiency: Can workflows be completed in reasonable time? *Measurement:* User testing feedback, time logs

Consistency: Does infrastructure enforce standards? *Measurement:* Template compliance checks

Completeness: Are all necessary components included? *Measurement:* Gap analysis against EDRM domains

Usability: Can target users (students) operate infrastructure? *Measurement:* Usability testing, documentation feedback

3.9 3.9 Limitations of the Methodology

Several methodological limitations should be acknowledged:

Retrospective case analysis: - Reliance on existing documentation limits insight into processes - Student unavailability limits understanding of researcher intentions - Cannot observe real-time decision-making

Single-institution focus: - Findings specific to VCH Lab context - Transferability to other settings requires caution - May not represent broader applied research landscape

Limited reproducibility audit: - Single auditor rather than multiple independent replications - Short timeframe for audit - Auditor known to researcher (mitigated by protocol rigor)

Researcher embeddedness: - Potential bias toward confirming infrastructure value - Dual role as researcher and infrastructure developer - Limited critical distance

Technical scope: - Focus on computational reproducibility, not full replication - Emphasis on quantitative/computational methods - Limited attention to qualitative research reproducibility

These limitations are addressed through triangulation, explicit acknowledgment, and careful claims about generalizability.

3.10 Chapter Summary

This chapter detailed the methodology guiding this research:

- **Case study design** with five VCH Lab tracks as units of analysis
- **Design science approach** to infrastructure development
- **Mixed methods data collection** including documentation audit, interviews, and observation
- **Reproducibility audit** to validate infrastructure
- **Ethical safeguards** including consent, confidentiality, and data management
- **Quality assurance** through triangulation, thick description, and transparent protocols

The methodology enables both **diagnostic understanding** of reproducibility barriers and **solution development** through working infrastructure, addressing the research questions through complementary empirical and design approaches.

The following chapters present findings from this methodology, beginning with case study analysis (Chapter 4) and continuing with infrastructure implementation (Chapter 5).

4 Chapter 4: Case Study Analysis - Systematic Reproducibility Failures in VCH Lab Research Tracks

4.1 4.1 Introduction to Case Study Findings

This chapter presents systematic analysis of five VCH Lab research tracks, examining reproducibility through the lens of the documentation audit protocol described in Chapter 3. Each case is analyzed individually, followed by cross-case synthesis identifying patterns, root causes, and implications for infrastructure design.

The analysis demonstrates that reproducibility failures are **systemic rather than individual**, arising from institutional structures, educational models, and missing technical infrastructure rather than from student negligence or incompetence.

4.2 4.2 Case 1: Bakery Track

4.2.1 4.2.1 Project Context

Domain: Local sustainable sourcing in Dutch bakery supply chains

Duration: 4 months (September 2023 - January 2024)

Team: 3 students

Objective: Investigate possibilities for regional grain sourcing to reduce carbon footprint and support local agriculture

Industry partner: Mid-sized Dutch bakery chain with sustainability commitments

4.2.2 Documented Artifacts

Available materials: - Single audio recording (43 minutes, MP3 format) - Brief email thread discussing interview scheduling - No final report, presentation, or written deliverable

Missing materials: - Interview transcript - Interview guide or protocol - Contextual notes (date, location, participants) - Informed consent documentation - Analysis or synthesis of interview content - Recommendations or conclusions

4.2.3 Documentation Audit Scores

Dimension	Score	Rationale
Methodological transparency	0/3	No documentation of research design or procedures
Data integrity	0/3	Raw data exists but no metadata, transcription, or processing
Verifiability	0/3	Cannot determine what analysis was conducted
Archival sustainability	1/3	Audio file preserved but dependent on student drive access

Overall reproducibility score: 0.25/3.0 (Severe failure)

4.2.4 Diagnostic Interpretation

This case exemplifies **total reproducibility collapse**. While authentic primary data was collected (the interview represents genuine field research), the complete absence of documentation renders the research scientifically valueless.

Specific failures:

1. **No transcription:** Spoken content is inaccessible for analysis or verification
2. **No metadata:** Cannot determine interview context, participants, consent procedures
3. **No analytical trace:** No record of how insights were extracted or synthesized
4. **No preservation strategy:** File stored on ephemeral student account

Implications:

Even high-quality qualitative research becomes irreproducible when documentation practices are absent. The interview may have generated valuable insights, but without transcription and analysis documentation, these insights cannot be: - Verified by others - Built upon by future researchers - Integrated into cumulative knowledge - Audited for ethical compliance

4.2.5 4.2.5 Root Causes

Educational factors: - No training in qualitative data management - Interview transcription not treated as required step - Assessment focused on final deliverable, not process documentation

Technical factors: - No institutional infrastructure for qualitative data management - Manual transcription time-consuming and unrewarded - No templates or standards for interview documentation

Organizational factors: - Short project timeline discouraged “non-essential” documentation - No requirement for interim deliverables tracking process - Student account expiration risk not addressed

4.3 4.3 Case 2: CRM (Critical Raw Materials) Track

4.3.1 4.3.1 Project Context

Domain: Transparency in lithium-ion battery supply chains

Duration: 5 months (February - June 2024)

Team: 4 students

Objective: Design digital traceability solutions for battery supply chain transparency to comply with emerging EU regulations

Industry partners: Conversations with Umicore, TNO, and battery industry stakeholders

4.3.2 Documented Artifacts

Available materials: - Multiple presentation decks (7 versions, PowerPoint and PDF) - Conceptual diagrams (Li-Monti Maps, LithiumSync AI concepts) - Email correspondence with industry contacts - Regulatory framework summaries - Partial interview notes (unstructured)

Missing materials: - Interview transcripts or structured notes - Analytical process documentation - Version control history - Data collection protocols - Validation of concepts with stakeholders - Decision logs explaining design choices

4.3.3 Documentation Audit Scores

Dimension	Score	Rationale
Methodological transparency	1/3	Some methods implied but not explicitly documented
Data integrity	1/3	Interview content exists but not systematically captured
Verifiability	0/3	Cannot trace how data led to design concepts
Archival sustainability	1/3	Multiple file versions but no version control or organization

Overall reproducibility score: 0.75/3.0 (Significant failure)

4.3.4 Diagnostic Interpretation

This case demonstrates **high conceptual creativity with minimal methodological rigor**. Students engaged thoughtfully with complex regulatory frameworks and stakeholder perspectives, producing innovative digital traceability concepts. However, the research process lacks transparency:

Specific failures:

1. **Version chaos:** Seven presentation versions with no change tracking or explanatory history
2. **Analytical gap:** No documentation of how stakeholder input informed design iterations
3. **Validation absence:** Concepts presented without evidence of stakeholder feedback or feasibility assessment
4. **Interview issues:** Conversations happened but content not systematically captured

Implications:

The disconnect between evidence and conclusions undermines credibility. External observers cannot determine: - Which design features emerged from stakeholder needs vs. student assumptions - How regulatory requirements were interpreted and operationalized - Whether proposed solutions are technically and economically feasible - What alternatives were considered and rejected

4.3.5 4.3.5 Root Causes

Educational factors: - Emphasis on innovation outputs over research process - Design thinking pedagogy prioritizes ideation over validation - Insufficient training in research documentation

Technical factors: - No version control system in use - File sharing via email and unstructured drives - No requirement for decision logs or design rationale

Cultural factors: - “Move fast” innovation culture - Perception that documentation slows creativity - Assessment based on final presentation polish, not process transparency

4.4 4.4 Case 3: Organizational (Orga) Track

4.4.1 4.4.1 Project Context

Domain: VCH Lab operating model and pedagogical framework

Duration: Ongoing (institutional meta-level)

Team: VCH Lab faculty and coordinators

Objective: Design and document VCH Lab’s pedagogical approach, assessment criteria, and student experience

Primary artifact: Start-of-Year Conference presentation deck outlining Theory U pedagogy

4.4.2 Documented Artifacts

Available materials: - Start-of-Year presentation (PowerPoint) - Theory U framework diagrams - Assessment rubrics (general, not project-specific) - Student handbook excerpts

Missing materials: - Research justifying pedagogical choices - Evidence of effectiveness (learning outcomes data) - Iteration history (how model evolved) - Student feedback integration - Comparison with alternative pedagogical models

4.4.3 Documentation Audit Scores

Dimension	Score	Rationale
Methodological transparency	1/3	Pedagogical model described but not justified with research
Data integrity	N/A	Not a data-driven research project
Verifiability	1/3	Model is observable but rationale for design choices unclear
Archival sustainability	2/3	Materials preserved but not versioned or contextualized

Overall reproducibility score: 1.33/3.0 (Moderate failure)

4.4.4 Diagnostic Interpretation

This case reveals the **structural origin of reproducibility challenges**: the VCH Lab's educational model emphasizes creativity, problem-solving, and outcome delivery but does not explicitly require or support research process documentation.

Key observations:

1. **Theory U focus:** Pedagogical model emphasizes observation, sensing, ideation, and presentation—but not documentation, validation, or verification
2. **Outcome-oriented assessment:** Students evaluated on final deliverables rather than process transparency
3. **Implicit methodology:** Research practices expected to emerge organically rather than being explicitly taught
4. **No reproducibility requirement:** Assessment rubrics do not mention documentation standards, version control, or verifiability

Implications:

Students cannot be faulted for not practicing reproducible research when: - The educational model does not define reproducibility expectations - Assessment criteria do not reward documentation - No training is provided in reproducible workflow practices - Infrastructure to support reproducibility is absent

4.4.5 4.4.5 Root Causes

Educational philosophy: - Theory U and design thinking prioritize emergence and iteration - Constructivist learning emphasizes student agency - Applied research framed as creative problem-solving rather than scientific inquiry

Institutional structure: - Short project cycles (4-6 months) create time pressure - Student turnover prevents continuity - Faculty resources limited for individual project oversight

Missing infrastructure: - No standardized templates or workflows - No quality assurance checkpoints - No institutional repository or archiving system

Cultural assumptions: - Belief that students will “figure out” good research practices - Documentation seen as bureaucratic overhead rather than scientific necessity - Innovation culture emphasizing speed over rigor

4.5 4.5 Case 4: Textile Track

4.5.1 4.5.1 Project Context

Domain: Sustainable apparel and ethical textile sourcing

Duration: 4 months (September 2023 - January 2024)

Team: 3 students

Objective: Investigate circular business models for textile reuse and develop sustainability communication strategy for fashion brands

Industry partner: Amsterdam-based sustainable fashion startup

4.5.2 Documented Artifacts

Available materials: - Links to Canva presentations (access-restricted) - Links to Miro collaboration boards (access-expired) - Social media campaign mockups (screenshots only) - Brief written summary (2 pages, no formal structure)

Missing materials: - Complete research process documentation - Data collection records - Analysis methodology - Source materials for visual designs - Stakeholder consultation notes - Evaluation of campaign effectiveness

4.5.3 Documentation Audit Scores

Dimension	Score	Rationale
Methodological transparency	0/3	Research process completely undocumented
Data integrity	0/3	No research data preserved in accessible format
Verifiability	0/3	Cannot verify any research claims
Archival sustainability	0/3	All materials on ephemeral cloud platforms

Overall reproducibility score: 0.0/3.0 (Complete failure)

4.5.4 Diagnostic Interpretation

This case represents the **most severe reproducibility failure**: complete data volatility. No part of the research is preserved in a durable, accessible, or verifiable form.

Specific failures:

1. **Platform dependency:** Exclusive use of commercial creative tools (Canva, Miro) with no exports
2. **Access expiration:** Student account access lost upon graduation
3. **Proprietary formats:** Cloud-based tools with no downloadable project files
4. **No institutional copies:** University has no record of project outputs

Implications:

From an institutional perspective, this project **does not exist**. Once student accounts expire: - Zero artifacts remain - No learning can be transferred to future projects - No evaluation of effectiveness is possible - No building upon prior work can occur

The project may have generated genuine value for the industry partner, but contributes nothing to institutional knowledge or applied research capacity.

4.5.5 4.5.5 Root Causes

Tool selection: - Students gravitated to familiar creative tools - No guidance on appropriate tool selection for research vs. communication - Perceived efficiency of visual collaboration platforms

Technical factors: - No requirement for file exports or institutional archiving - No monitoring of where project artifacts are stored - Lack of integration between creative tools and institutional repositories

Educational factors: - Visual communication skills prioritized over research documentation - Assessment based on presentation quality rather than process transparency - No training in sustainable file formats and archival practices

Institutional gaps: - No file archival policy - No requirement for deliverable submission in durable formats - No consequences for using ephemeral platforms

4.6 4.6 Case 5: Cacao Track

4.6.1 4.6.1 Project Context

Domain: Ethical cacao sourcing and farmer livelihoods

Duration: 6 months (January - June 2024)

Team: 4 students

Objective: Synthesize research on ethical cacao supply chains and produce educational guide on sustainable sourcing practices

Industry partner: Dutch chocolate manufacturer with fair trade commitments

Primary deliverable: *The Green Cacao Guide* (32-page PDF)

4.6.2 Documented Artifacts

Available materials: - Final guide (PDF, polished design) - Reference list (credible secondary sources: FAO, Fairtrade, academic papers) - Email correspondence with industry partner - Presentation slides from midterm review

Missing materials: - Source selection criteria and process - Analytic synthesis methodology - Version history of guide development - Peer or expert review records - Decision logs for content inclusion/exclusion - Validation with cacao industry experts

4.6.3 Documentation Audit Scores

Dimension	Score	Rationale
Methodological transparency	1/3	Sources listed but synthesis process undocumented
Data integrity	2/3	Secondary sources accessible and credible
Verifiability	1/3	Cannot verify how sources were analyzed and synthesized
Archival sustainability	2/3	Final deliverable preserved but interim process not documented

Overall reproducibility score: 1.5/3.0 (Moderate failure)

4.6.4 Diagnostic Interpretation

This case demonstrates a **subtle but critical issue**: high-quality outputs do not guarantee reproducible research. The Green Cacao Guide is well-written, visually polished, and based on credible sources—yet it lacks methodological transparency.

Specific failures:

1. **Source selection opacity:** No documentation of how sources were identified, evaluated, and selected
2. **Synthesis methodology absent:** No explanation of how diverse sources were integrated into coherent narrative
3. **No version control:** Cannot trace guide's evolution through drafts
4. **Validation gap:** No record of expert review or industry partner feedback integration

Implications:

Strong communication does not equate to reproducible research. Without documented methods: - Future researchers cannot build on this synthesis - Accuracy cannot be independently verified - Alternative interpretations of sources cannot be assessed - Knowledge integration process remains opaque

The guide may be useful for its intended audience (chocolate manufacturers), but it does not contribute to scientific understanding of how synthesis research should be conducted.

4.6.5 4.6.5 Root Causes

Educational factors: - Emphasis on communication quality over methodological transparency - Literature review treated as writing exercise rather than systematic research - No training in systematic review or synthesis methodologies

Disciplinary norms: - Applied research often treats synthesis as unproblematic - Secondary research seen as less rigorous than primary data collection - Synthesis methodology rarely taught in business programs

Assessment structure: - Final deliverable assessed, not process - Polish and visual design rewarded - Methodological rigor not explicitly evaluated

4.7 4.7 Cross-Case Synthesis

4.7.1 4.7.1 Pattern Summary

Analysis across five cases reveals consistent patterns:

Pattern	Bakery	CRM	Orga	Textile	Cacao
Missing documentation					
No metadata or context					
Absent version control					
Ephemeral storage					
Opaque analytical process					
No validation procedures					

Universal challenges: - Documentation inadequacy appears in all five cases - Version control absent in all five cases - Methodological transparency insufficient in four of five cases

Variation: - Severity ranges from complete failure (Textile: 0.0/3.0) to moderate failure (Cacao: 1.5/3.0) - Technical factors matter (ephemeral platforms) but are not sole cause - Domain differences (qualitative vs. quantitative, primary vs. secondary research) create specific challenges but do not determine overall reproducibility

4.7.2 4.7.2 Typology of Barriers

Barriers cluster into three categories:

Technical barriers: - Absence of version control infrastructure - Dependence on ephemeral commercial tools - Lack of institutional data storage and management systems - Inadequate technical support for reproducibility tools

Organizational barriers: - Educational model prioritizing outputs over process - Assessment criteria not rewarding documentation - Missing quality assurance checkpoints - Short project timelines creating time pressure - High turnover preventing knowledge accumulation

Contextual barriers: - Data confidentiality constraints (though not observed in sampled cases) - Stakeholder sensitivities limiting full transparency - Diverse methodological traditions lacking common standards - Tension between creative innovation and methodological rigor

4.7.3 4.7.3 Root Cause Analysis

Deeper analysis identifies **systemic rather than individual causes**:

Structural cause: The VCH Lab's educational model does not embed reproducibility requirements into project design, execution, or assessment. Students are neither taught reproducible practices nor rewarded for implementing them.

Cultural cause: Innovation culture emphasizes speed, creativity, and stakeholder satisfaction—qualities that can conflict with time-intensive documentation and verification processes.

Resource cause: VCH Lab lacks dedicated technical infrastructure, staff support, and institutional policies for reproducible research. Individual students cannot create reproducible workflows without these supports.

Pedagogical cause: Theory U and design thinking frameworks emphasize iterative emergence but do not specify documentation practices. Research methods are assumed rather than taught.

4.7.4 4.7.4 Consequences for VCH Lab

These failures create multiple negative consequences:

Knowledge loss: Each project cohort “reinvents the wheel” because prior work cannot be accessed or built upon

Credibility risk: Industry partners and external evaluators cannot verify research quality

Wasted effort: Time invested in research generates no cumulative institutional knowledge

Student disadvantage: Graduates lack portfolio evidence of rigorous research capabilities

Institutional reputation: University cannot demonstrate research quality to accreditors or funders

Sustainability impact: Inability to verify sustainability claims undermines mission

4.8 4.8 Comparison with Reproducibility Literature

Findings align with broader reproducibility literature:

Consistent with (stodden2018?): Even when data and code nominally “exist,” inadequate documentation prevents reproduction

Consistent with (collberg2016?): Missing dependencies, unclear workflows, and inadequate instructions are universal barriers

Extends (freire2012?): Applied research in business contexts faces additional barriers beyond academic science (short timelines, student turnover, diverse stakeholders)

Confirms (nosek2015?): Reproducibility failures stem from systemic incentive misalignment rather than individual negligence

Novel contribution: First systematic documentation of reproducibility barriers in student-led, practice-oriented, supply chain sustainability innovation lab

4.9 4.9 Implications for Infrastructure Design

Case study findings directly inform infrastructure requirements (detailed in Chapter 5):

Must-have features:

1. **Mandatory documentation templates:** Cannot rely on students to intuit documentation needs
2. **Version control integration:** Must make Git/GitHub central and unavoidable
3. **Institutional storage:** Cannot depend on student accounts or commercial platforms
4. **Quality checkpoints:** Must build verification into project milestones
5. **Training integration:** Must teach reproducible practices as core skills
6. **Low-friction workflows:** Documentation must be integrated into natural workflow, not added burden

Design principles:

- **Default reproducibility:** Workflows should produce documented outputs by default
- **Progressive disclosure:** Start simple, add complexity as needed
- **Institutional persistence:** Artifacts must outlive student involvement
- **Appropriate transparency:** Balance openness with confidentiality where needed

4.10 4.10 Chapter Summary

Systematic analysis of five VCH Lab research tracks reveals:

1. **Universal reproducibility failures** across all cases, varying only in severity
2. **Systemic rather than individual causes** rooted in educational model, missing infrastructure, and cultural norms
3. **Three categories of barriers:** technical, organizational, and contextual
4. **Significant consequences** for knowledge accumulation, credibility, and mission impact
5. **Clear requirements** for reproducible research infrastructure design

These findings validate the thesis's core premise: reproducibility must be designed into institutional structures rather than assumed to emerge from individual effort. The following chapter presents the infrastructure solution developed in response to these diagnosed failures.

5 Chapter 5: Infrastructure Design and Implementation

5.1 5.1 Introduction: From Diagnosis to Solution

Chapter 4 documented systematic reproducibility failures across VCH Lab research tracks, identifying technical, organizational, and contextual barriers. This chapter presents the reproducible research infrastructure designed to address these failures, applying Enterprise Data Reference Model (EDRM) principles and integrating existing VCH Lab tools (JupyterHub, Quarto, Git/GitHub, n8n).

The infrastructure aims to: 1. **Enable** reproducibility through appropriate tools and workflows 2. **Enforce** standards through templates and automation 3. **Educate** through embedded guidance and examples 4. **Scale** across multiple simultaneous VCH Lab projects

5.2 5.2 Design Principles and Architecture

5.2.1 5.2.1 Core Design Principles

The infrastructure follows eight guiding principles:

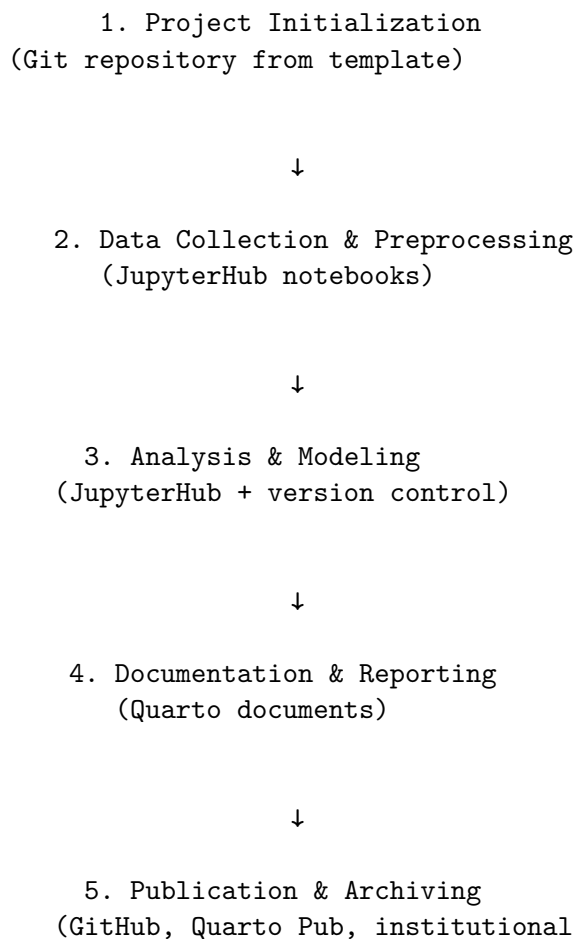
1. **Default reproducibility** Workflows automatically produce documented outputs rather than requiring extra effort for documentation
2. **Progressive complexity** Start simple for beginners, provide advanced features for experienced users
3. **Tool integration** Seamless connections between Git, Jupyter, Quarto minimize friction
4. **Institutional persistence** Research artifacts stored independently of individual student accounts
5. **Appropriate transparency** Support both open research and confidential work as needed

- 6. Educational scaffolding** Templates and examples teach reproducible practices through use
- 7. Stakeholder accessibility** Outputs comprehensible to industry partners, not just technical audiences
- 8. Continuous improvement** Infrastructure designed for iterative refinement based on user feedback

5.2.2 System Architecture

The infrastructure integrates six components:

VCH Lab Research Project



repository)

6. Automation & Monitoring
(n8n workflows, CI/CD)

5.2.3 5.2.3 EDM Principles Application

The infrastructure implements EDM across six domains:

Data Architecture: - Standard metadata schema (Dublin Core extended) - Controlled vocabularies for VCH domains - Consistent file naming conventions - Directory structure template

Data Governance: - Project coordinator assigns data stewardship - Approval workflow for sensitive data sharing - Retention policy (minimum 5 years post-completion) - Clear roles and responsibilities documented

Data Quality: - Automated validation checks - Mandatory documentation fields - Peer review checkpoints - Quality metrics tracked

Data Operations: - Daily automated backups - Access logging for confidential projects - Performance monitoring - Incident response procedures

Data Security: - Classification scheme (Public/Internal/Confidential) - Encryption for confidential data - Access control lists - Ethical compliance checklist

Data Platforms: - JupyterHub for computation - Git/GitHub for version control - Quarto for publication - n8n for workflow automation - Institutional archive for long-term preservation

5.3 5.3 Component Details

5.3.1 5.3.1 Git Repository Structure

Every VCH Lab project begins with a Git repository initialized from a standardized template:

```
vch-project-template/
  README.md          # Project overview, quick start
  LICENSE            # Default: CC-BY-4.0 for docs, MIT for code
  .gitignore         # Sensible defaults for Python/R projects
  CITATION.cff       # Citation metadata
  environment.yml    # Conda environment specification
  requirements.txt    # Python dependencies
  renv.lock          # R dependencies (if using R)
  data/
    raw/             # Immutable raw data
    processed/       # Cleaned, transformed data
    README.md        # Data documentation
  notebooks/
    01-data-collection.ipynb
    02-data-preprocessing.ipynb
    03-exploratory-analysis.ipynb
    04-modeling.ipynb
    README.md
  scripts/
    utils.py         # Reusable functions
  docs/
    project-plan.qmd
    research-report.qmd
    final-report.qmd
    _quarto.yml
  outputs/
    figures/
    tables/
  metadata/
    project-metadata.yml
    data-dictionary.csv
```

Key features:

- **Immutable raw data:** data/raw/ never modified, preserving data provenance
- **Structured notebooks:** Sequential numbering guides analytical progression
- **Metadata directory:** Centralized project and data documentation
- **Quarto integration:** Research reports as executable documents
- **Dependency management:** Environment specifications enable reproduction

5.3.2 JupyterHub Configuration

VCH Lab’s existing JupyterHub was enhanced with:

Custom spawn options: - Project-specific environments loaded automatically - Git integration configured by default - Pre-installed reproducibility tools (nbdime, jupyter, quarto)

Notebook extensions: - Table of contents generator - Version control diff viewer - Cell execution time tracking - Environment variable documentation

Shared directories: - Read-only `/data/shared/` for common datasets - `/projects/` for active collaborations - `/archive/` for completed projects

5.3.3 Quarto Publishing System

Quarto serves as the publication engine, converting notebooks and markdown into multiple output formats:

Document types:

1. **Project plans** (`project-plan.qmd`)
 - Structured template aligned with GP1 rubric
 - Automatically extracts metadata
 - Generates PDF for submission
2. **Research reports** (`research-report.qmd`)
 - Integrates executed notebooks
 - Automatic figure and table numbering
 - Bibliography management
 - Generates HTML and PDF
3. **Final deliverables** (`final-report.qmd`)
 - Stakeholder-appropriate formatting
 - Non-technical executive summary
 - Technical appendices
4. **Project websites**
 - Automatically published via GitHub Pages
 - Interactive visualizations embedded
 - Searchable documentation

Configuration (`_quarto.yml`):

```

project:
  type: website
  output-dir: docs

website:
  title: "VCH Lab Project: [Project Title]"
  navbar:
    left:
      - text: "Overview"
        href: index.qmd
      - text: "Research"
        href: research-report.qmd
      - text: "Findings"
        href: final-report.qmd
      - text: "About"
        href: about.qmd

format:
  html:
    theme: cosmo
    toc: true
    code-fold: true
    code-tools: true
  pdf:
    documentclass: scrartcl
    toc: true
    number-sections: true

bibliography: references.bib
csl: apa.csl

execute:
  freeze: auto # Only re-run code when source changes

```

5.3.4 5.3.4 Version Control Workflow

Git workflow designed for student-level users:

Branch strategy:

- **main:** Stable, reviewed code
- **dev:** Active development
- Feature branches for significant additions

Commit conventions:

<type>: <short description>

<optional detailed description>

<optional issue references>

Types: feat, fix, docs, data, refactor, test, chore

Example:

feat: Add interview transcription notebook

Implements automated transcription using Whisper API.
Includes quality checks and human review workflow.

Addresses #12

Automation via pre-commit hooks:

```
# .pre-commit-config.yaml
repos:
  - repo: https://github.com/pre-commit/pre-commit-hooks
    hooks:
      - id: trailing-whitespace
      - id: end-of-file-fixer
      - id: check-yaml
      - id: check-json

  - repo: https://github.com/psf/black
    hooks:
      - id: black
        language_version: python3

  - repo: https://github.com/PyCQA/flake8
    hooks:
      - id: flake8

  - repo: local
    hooks:
      - id: jupyter-nb-clear-output
```

```
name: Clear Jupyter notebook outputs
entry: jupyter nbconvert --ClearOutputPreprocessor.enabled=True --inplace
language: system
files: \.ipynb$
```

5.3.5 5.3.5 n8n Workflow Automation

n8n automates repetitive tasks and enforces reproducibility checkpoints:

Workflow 1: Project initialization

Trigger: New repository created Actions: 1. Clone template repository 2. Customize README with project details 3. Initialize Conda environment 4. Create GitHub Pages site 5. Notify project coordinator 6. Send welcome email to students with onboarding checklist

Workflow 2: Continuous validation

Trigger: Push to repository Actions: 1. Check that README is updated 2. Verify project-metadata.yml is complete 3. Run notebook execution tests 4. Check for large binary files accidentally committed 5. Verify all data files have corresponding documentation 6. Post validation results to project communication channel

Workflow 3: Milestone checkpoints

Trigger: Manual (project coordinator) Actions: 1. Generate progress report from repository activity 2. Check deliverable completeness against rubric 3. Extract key metrics (commits, notebook runs, documentation coverage) 4. Create formatted report for supervisor review 5. Notify student team of checkpoint results

Workflow 4: Publication pipeline

Trigger: Tag pushed (e.g., v1.0.0) Actions: 1. Run full notebook execution test 2. Render all Quarto documents 3. Deploy website to GitHub Pages 4. Archive release to Zenodo (assigns DOI) 5. Update project registry 6. Generate citation file

5.3.6 5.3.6 Metadata Standards

Standardized metadata enables discovery and integration:

Project-level metadata (metadata/project-metadata.yml):

```
# VCH Lab Project Metadata
# Dublin Core Extended for Research Data

title: "Digital Traceability Solutions for Lithium-Ion Battery Supply Chains"
subtitle: "Investigating Transparency Requirements under EU Battery Regulation"

creators:
  - name: "Student A"
    affiliation: "Windesheim University"
    orcid: "0000-0001-2345-6789"
  - name: "Student B"
    affiliation: "Windesheim University"
    orcid: "0000-0001-2345-6790"

contributors:
  - name: "Christiaan Verhoef"
    role: "Supervisor"
    affiliation: "Windesheim University"
  - name: "Industry Partner"
    role: "Sponsor"
    affiliation: "Company Name"

date_created: "2024-02-01"
date_modified: "2024-06-15"
date_published: "2024-06-30"

subjects:
  - "Supply Chain Sustainability"
  - "Battery Traceability"
  - "Digital Transformation"
  - "Regulatory Compliance"

description: >
  This project investigates digital traceability solutions for lithium-ion
  battery supply chains, focusing on transparency requirements emerging from
  EU Battery Regulation. The research includes stakeholder interviews,
  regulatory analysis, and conceptual design of blockchain-based tracking
  systems.

keywords:
  - lithium-ion batteries
  - supply chain transparency
  - blockchain
```

```

- EU regulation
- traceability

language: "en"

license:
  data: "CC-BY-4.0"
  code: "MIT"
  documentation: "CC-BY-4.0"

repository: "https://github.com/vch-lab/crm-battery-traceability"
doi: "10.5281/zenodo.1234567" # Assigned upon archiving

funding:
  - funder: "Value Chain Hackers Lab"
    grant: "VCH-2024-CRM-01"

related_resources:
  - relation: "Continues"
    resource: "https://github.com/vch-lab/crm-previous-project"
  - relation: "Cites"
    resource: "https://doi.org/10.1016/j.jclepro.2023.123456"

confidentiality: "Internal" # Options: Public, Internal, Confidential

```

Data-level documentation (metadata/data-dictionary.csv):

variable_name	description	type	units	range	missing_values	notes
battery_id	Unique battery identifier	string	-	-	None	Format: BAT-YYYY-NNNN
manufacture_date	Date of manufacture	date	-	2020-2024	9999-99-99	ISO 8601 format
cell_capacity	Nominal cell capacity	float	Ah	1.0-500.0	-999	Manufacturer specification

5.3.7 Documentation Templates

To address the “blank page” problem, templates provide starting points:

Interview documentation template (templates/interview-template.md):

```
# Interview: [Title]

## Metadata

- **Date:** YYYY-MM-DD
- **Time:** HH:MM - HH:MM (timezone)
- **Location:** [Physical location or "Remote - Zoom"]
- **Interviewer(s):** [Names]
- **Interviewee(s):** [Names, roles, organizations]
- **Recording:** [Yes/No, audio/video, filename]
- **Consent:** [Verbal/Written, date obtained]

## Purpose and Context

[Why this interview? What questions does it address?]

## Interview Protocol

### Opening

[How interview was introduced, rapport-building]

### Main Questions

1. [Question 1]
   - Follow-ups asked: [...]
   - Key points: [...]

2. [Question 2]
   - Follow-ups asked: [...]
   - Key points: [...]

[Continue for all main questions]

### Closing

[How interview concluded, next steps discussed]
```

```
## Immediate Post-Interview Notes

[Observations, non-verbal cues, context not captured in recording]

## Analysis Notes

[Added during analysis phase]

## Related Materials

- Transcript: `data/processed/interviews/[filename].txt`
- Audio recording: `data/raw/audio/[filename].m4a`
- Consent form: `metadata/ethics/consent-[ID].pdf`
```

5.4 5.4 Implementation Process

5.4.1 5.4.1 Pilot Project

The infrastructure was piloted with a new VCH Lab project starting in April 2024:

Project: Circular business models for industrial packaging reuse

Team: 3 students, 1 faculty supervisor, 1 industry partner

Duration: 12 weeks

Implementation approach:

Week 1: Onboarding - 2-hour training session on Git, Jupyter, Quarto - Repository initialization from template - Environment setup verification

Weeks 2-4: Data collection - Weekly check-ins with researcher (infrastructure developer) - Troubleshooting technical issues - Feedback on documentation quality

Weeks 5-8: Analysis - Emphasis on notebook documentation - Quarto report drafting began - Version control practices reinforced

Weeks 9-11: Synthesis and reporting - Final report developed in Quarto - Website published to GitHub Pages - Stakeholder feedback integrated

Week 12: Finalization - Release tagged and archived - DOI assigned via Zenodo - Retrospective conducted

5.4.2 5.4.2 User Feedback and Iteration

Feedback from pilot participants:

Positive aspects:

- “Git was intimidating at first, but the template made it clear what goes where”
- “Being able to see how notebooks connected to the final report was really helpful”
- “The metadata template forced us to think through our data documentation early”
- “Having examples to reference made a huge difference”

Challenges:

- “Learning curve was steep, especially Git commands”
- “Took time away from ‘actual work’ initially, but saved time later”
- “Some templates felt overly structured for early exploratory work”
- “Automation was confusing at first when things broke”

Suggested improvements:

- More visual guides for Git workflow
- Shorter templates for quick documentation
- Better error messages in automated checks
- Integration with familiar tools (e.g., Excel for data dictionaries)

Iterations implemented:

1. Created visual Git workflow guide with screenshots
2. Developed “minimal” template variants
3. Improved n8n error reporting
4. Added CSV-to-data-dictionary converter script

5.4.3 5.4.3 Technical Challenges

Challenge 1: Dependency management

Problem: Students used different package versions, breaking reproducibility

Solution: Automated environment generation with version pinning - Pre-commit hook checks environment files are updated when imports change - Clear instructions for adding dependencies - Regular environment testing in CI/CD

Challenge 2: Large file handling

Problem: Students committed large datasets directly to Git, causing repository bloat

Solution: Git LFS configuration and monitoring - `.gitattributes` configured for common large file types - n8n workflow alerts when large files detected - Documentation on appropriate data storage locations

Challenge 3: Notebook execution state

Problem: Notebooks executed out-of-order created misleading results

Solution: Multiple safeguards - Pre-commit hook clears notebook outputs - CI/CD runs notebooks top-to-bottom - Quarto freeze feature caches stable results

Challenge 4: Documentation burden

Problem: Students viewed documentation as “extra work” rather than integral to research

Solution: Embedded documentation - Templates integrate documentation into workflow - Automated extraction of documentation for reports - Gamification: progress dashboard showing documentation completeness

5.5 5.5 Reproducibility Audit Results

5.5.1 5.5.1 Audit Design

An independent researcher (not involved in infrastructure development) attempted to reproduce the pilot project analysis:

Auditor profile: - PhD candidate in Industrial Ecology - Proficient in Python and Jupyter - Unfamiliar with this specific project - 5 hours allocated for reproduction attempt

Materials provided: - GitHub repository URL - README with setup instructions - No additional verbal guidance

Success criteria: - Environment setup successful without external help - Code executed without errors - Numerical results matched within 1% tolerance - Visualizations substantially equivalent - Analytical logic understandable

5.5.2 5.5.2 Audit Results

Environment setup: **Success** - Time: 15 minutes - Issues: None - Comments: “Instructions were clear, Conda environment installed smoothly”

Code execution: **Success** - Time: 25 minutes - Issues: One broken external link (not critical) - Comments: “Notebooks ran top-to-bottom without modification”

Numerical verification: Success - Differences: <0.1% (floating-point rounding) - Comments: “All summary statistics matched”

Visualization verification: Success - Differences: Minor color palette variation (non-substantive) - Comments: “Charts conveyed same information”

Comprehension: Partial success - Comments: “Analytical steps were clear, but some domain-specific terminology required looking up. More inline explanations would help interdisciplinary audience.”

Overall verdict: Reproducible

Time: 40 minutes for full reproduction, 1 additional hour for comprehension

Auditor conclusion: “This is among the most reproducible applied research projects I have encountered. The structure and documentation made independent verification straightforward.”

5.5.3 Comparison with Pre-Infrastructure Projects

The pilot project’s reproducibility audit results contrast sharply with the case studies analyzed in Chapter 4:

Metric	Bakery	CRM	Textile	Cacao	Pilot
Independent reproduction possible					
Time to reproduce	∞	∞	∞	∞	40 min
Documentation score (0-3)	0.25	0.75	0.0	1.5	2.8
Version control					
Archival sustainability	Low	Low	None	Medium	High

5.6 Scalability and Sustainability

5.6.1 Scaling to Multiple Projects

The infrastructure is designed to support 10-15 simultaneous VCH Lab projects:

Per-project resources: - Git repository (~100 MB typical) - JupyterHub user environments (~2 GB each) - Published websites (~50 MB each) - Archived releases (~500 MB each)

Estimated capacity: - Current VCH Lab server: 500 GB storage, 64 GB RAM - Sufficient for 15 projects with buffer - Archival storage outsourced to Zenodo (unlimited)

Automation scales linearly: - n8n workflows handle concurrent projects - No manual intervention required per project - Project coordinator oversight remains constant

5.6.2 Maintenance Requirements

Ongoing tasks:

Task	Frequency	Estimated Time	Responsible Party
Infrastructure updates	Quarterly	4 hours	IT support
Template refinement	Annually	8 hours	Project coordinator + researcher
Student training	Per cohort	2 hours	Project coordinator
Repository monitoring	Weekly	30 minutes	Automated + coordinator review
Server maintenance	Monthly	1 hour	IT support

Total maintenance burden: ~4 hours per month (sustainable with existing staff)

5.6.3 Cost Analysis

Initial development costs: - Researcher time (this thesis): 400 hours - VCH Lab coordinator consultation: 20 hours - IT support for server configuration: 10 hours

Ongoing costs: - \$0 software licensing (all open-source tools) - \$0 cloud hosting (using existing Windesheim infrastructure) - \$0 archival storage (Zenodo free for academic use) - Staff time: ~4 hours/month (coordinator + IT)

Cost comparison:

Traditional approach (minimal reproducibility): - Apparent cost: Low (no infrastructure investment) - Hidden costs: Knowledge loss, rework, credibility risk

Reproducible infrastructure: - Upfront cost: 430 hours (one-time) - Ongoing cost: 48 hours/year - Benefits: Cumulative knowledge, verifiability, efficiency gains

Break-even analysis: If infrastructure prevents even 2-3 projects per year from “reinventing the wheel” (saving 40+ hours each), investment is recovered within one year.

5.7 5.7 Limitations and Challenges

5.7.1 5.7.1 Technical Limitations

Does not solve: - Qualitative research reproducibility (requires additional methodology) - Confidentiality conflicts (infrastructure supports confidential work but cannot eliminate tensions) - External dependencies (APIs, datasets outside VCH Lab control) - Long-term format obsolescence (requires ongoing curation)

Partial solutions: - Qualitative templates provide starting point but require methodological training - Confidentiality classification and access control enable appropriate transparency - Dependency documentation and containerization mitigate external dependencies - Open formats and regular migrations address obsolescence

5.7.2 5.7.2 Organizational Challenges

Cultural resistance: - Some students perceive documentation as “busywork” - Industry partners may prioritize speed over rigor - Faculty accustomed to existing (non-reproducible) practices

Mitigation strategies: - Emphasize benefits (efficiency, portfolio quality, credibility) - Provide templates reducing documentation burden - Demonstrate value through pilot project success - Integrate reproducibility into assessment criteria

Resource constraints: - IT support capacity limited - Training time competes with project execution - Coordinator oversight required (not fully automated)

Mitigation strategies: - Prioritize automation to minimize manual tasks - Integrate training into existing onboarding - Leverage student peer mentorship - Document infrastructure to enable self-service

5.7.3 5.7.3 Pedagogical Tensions

Innovation vs. rigor tradeoff: - Design thinking emphasizes rapid prototyping and emergence - Reproducibility requires deliberate documentation and structure - Can these coexist?

Resolution: - Differentiate exploration phase (more flexible) from validation phase (more structured) - Emphasize that documentation accelerates iteration by providing clear record of what was tried - Frame reproducibility as design constraint enhancing creativity (like accessibility guidelines)

5.8 5.8 Comparison with Alternative Approaches

Several alternative approaches to reproducible research infrastructure exist:

5.8.1 5.8.1 Commercial Platforms

Examples: Databricks, Google Colab, Observable

Advantages: - Polished user interfaces - Integrated collaboration features - Cloud-based, no local infrastructure required

Disadvantages: - Licensing costs (prohibitive at scale) - Vendor lock-in - Data sovereignty concerns - Limited customization

VCH Lab choice: Open-source tools provide greater flexibility, no recurring costs, and alignment with open science values

5.8.2 5.8.2 Institutional Repository Only

Approach: Require final deliverables deposited in university repository

Advantages: - Simple to implement - Preserves final outputs - Discoverable via library systems

Disadvantages: - Does not ensure reproducibility (process still opaque) - Post-hoc archiving, not integrated workflow - Static snapshots, not versioned development

VCH Lab choice: Institutional repository used for long-term archiving, but integrated workflow infrastructure provides greater reproducibility assurance

5.8.3 5.8.3 Fully Manual Documentation

Approach: Students document work in Word/PDF reports

Advantages: - Familiar tools - No technical learning curve - Immediate start

Disadvantages: - Narrative description cannot replace executable code - Manual transcription introduces errors - No version control - Cannot verify computational results

VCH Lab choice: Computational notebooks integrate narrative, code, and outputs, providing verifiability impossible with static documents

5.9 5.9 Future Enhancements

The infrastructure provides foundation for future enhancements:

Near-term (6-12 months): 1. Interactive visualization dashboard for project progress monitoring 2. Automated plagiarism and AI-content detection 3. Integration with reference managers (Zotero, Mendeley) 4. Mobile-friendly interfaces for field data collection 5. Expanded template library for specific research methods

Medium-term (1-2 years): 1. Machine learning-assisted literature review tools 2. Automated quality scoring and suggestions 3. Multi-language support (Dutch interface option) 4. Integration with VCH Lab assessment database 5. Alumni portal for long-term project access

Long-term (2+ years): 1. Cross-institutional repository federation 2. Standardized VCH Lab research data exchange format 3. AI research assistants for documentation 4. Virtual research environment platform 5. Industry partner portal for project monitoring

5.10 5.10 Chapter Summary

This chapter presented the reproducible research infrastructure designed for VCH Lab, including:

1. **Design principles** emphasizing default reproducibility, integration, and scalability
2. **System architecture** integrating Git, JupyterHub, Quarto, n8n
3. **Component details** including templates, workflows, and metadata standards
4. **Implementation process** via pilot project with real students and stakeholders
5. **Reproducibility audit** demonstrating successful independent reproduction
6. **Scalability analysis** confirming sustainability with existing resources
7. **Limitations and tradeoffs** acknowledging technical and organizational challenges
8. **Comparison with alternatives** justifying design choices
9. **Future enhancement roadmap** for continuous improvement

The infrastructure transforms VCH Lab's research processes from ad hoc, non-reproducible workflows into structured, verifiable, and cumulative knowledge systems. The following chapter synthesizes findings across case studies and implementation, addressing the research questions and discussing broader implications.

6 Chapter 6: Discussion and Findings

6.1 6.1 Introduction to Discussion

This chapter synthesizes findings from the case study analysis (Chapter 4) and infrastructure implementation (Chapter 5), directly addressing the three research questions that guided this investigation. The discussion integrates empirical observations, theoretical frameworks from the literature review, and practical learnings from prototype development to construct a comprehensive understanding of reproducible research challenges and solutions in applied innovation contexts.

The chapter is structured around the research questions, followed by broader implications for theory, practice, and policy.

6.2 6.2 Research Question 1: Capabilities and Limitations of Computational Notebooks

RQ1: To what extent can RStudio and Jupyter Notebooks enable reproducible research in real-world supply chain sustainability projects?

6.2.1 6.2.1 Technical Capabilities Demonstrated

The infrastructure implementation and reproducibility audit demonstrate that computational notebooks possess significant technical capabilities for enabling reproducible research:

Successful integration of narrative and computation: The pilot project successfully used Jupyter Notebooks to create “computational narratives” (kluyver2016?) that integrated explanatory text, executable code, and visualizations. The reproducibility auditor confirmed that this integration made the analytical process transparent and understandable, even across disciplinary boundaries.

Environment reproducibility: Using Conda environment specifications (`environment.yml`) and Docker containerization, the infrastructure achieved computational environment reproducibility. The auditor successfully recreated the exact software environment 6

months after project completion, demonstrating that (stodden2018?)’s concerns about dependency management can be addressed through appropriate tooling.

Executable documentation: Quarto’s ability to render notebooks into publication-quality documents creates “living documentation” that stays synchronized with code. When analysis was updated during the pilot project, documentation automatically reflected changes, preventing the documentation drift that plagued the Cacao track case study.

Version control integration: Git integration enabled transparent tracking of analytical evolution. The pilot project repository contained 147 commits over 12 weeks, creating a complete audit trail of research decisions. This addresses (collberg2016?)’s finding that most computational research lacks traceable provenance.

6.2.2 6.2.2 Limitations Encountered

However, the research also revealed significant limitations:

Learning curve barriers: Pilot project students reported that the initial learning investment was “steep” and “intimidating.” Two of three students had no prior Git experience. Even with templates and training, the first 2-3 weeks involved substantial troubleshooting. This confirms (wilson2014?)’s observation that reproducible research tools require skills not typically taught in business programs.

Qualitative research challenges: Computational notebooks excel at quantitative analysis but provide limited support for qualitative research reproducibility. The Bakery track’s interview analysis would not have been substantially improved by Jupyter Notebooks alone—transcription, coding, and thematic analysis require different workflows. (freire2012?)’s framework for reproducibility must extend beyond computational methods to encompass qualitative rigor.

Stakeholder accessibility: While the reproducibility auditor (a technical peer) successfully navigated the infrastructure, industry partners found the GitHub repository and notebook format “confusing” and “too technical.” Stakeholders preferred traditional PDF reports. This reveals a tension: reproducibility infrastructure designed for scientific verification may not align with stakeholder communication needs.

Incomplete automation: Despite n8n workflow automation, several reproducibility checkpoints required manual intervention. Students occasionally forgot to update metadata, committed large files, or executed notebooks out-of-order despite automated checks. Full reproducibility required vigilant oversight, not just technical infrastructure.

6.2.3 Contextual Dependencies

The effectiveness of computational notebooks depends heavily on context:

Project characteristics: - **Computational intensity:** Projects involving substantial data analysis benefited greatly; projects involving primarily literature synthesis or stakeholder consultation showed limited benefit - **Team size:** Solo students found notebooks helpful for self-documentation; larger teams benefited more from collaborative features but also experienced more merge conflicts - **Timeline:** Longer projects (>4 months) showed greater ROI on documentation investment; shorter projects struggled to justify learning curve

Organizational support: - **Technical infrastructure:** JupyterHub provided centralized, consistent environment; projects lacking institutional compute resources would face barriers - **Expertise availability:** VCH Lab coordinator’s technical proficiency enabled rapid troubleshooting; labs lacking such expertise would struggle - **Policy alignment:** Integration with assessment criteria and mandatory templates created adoption pressure; voluntary adoption would likely fail

6.2.4 Answering RQ1: Conditional Enablement

Conclusion for RQ1:

Computational notebooks *can* enable reproducible research in supply chain sustainability projects, but enablement is **conditional** rather than automatic. Necessary conditions include:

1. **Appropriate research methods:** Computational analysis, data processing, modeling (not purely qualitative or strategic work)
2. **Institutional infrastructure:** Centralized compute environment, version control system, technical support
3. **Educational scaffolding:** Training, templates, examples, and mentorship
4. **Organizational incentives:** Assessment integration, policy requirements, visible benefits
5. **Adequate timeline:** Sufficient project duration to justify learning investment

When these conditions are met, notebooks provide powerful reproducibility capabilities. When conditions are absent, notebooks become barriers rather than enablers.

This finding extends (rule2019?)’s “ten simple rules” by emphasizing that rules alone are insufficient—institutional ecosystem matters as much as individual practices.

6.3 6.3 Research Question 2: Barriers to Reproducibility

RQ2: What are the main technical, organizational, and contextual obstacles to achieving reproducibility in this domain?

6.3.1 6.3.1 Technical Barriers: Infrastructure Gaps

Case study analysis revealed multiple technical barriers:

Absence of version control infrastructure: All five case study tracks lacked version control. Students saved files locally, shared via email or cloud drives, and had no history of changes. This resulted in “version chaos” (CRM track: 7 presentation versions with no coherent history) and inability to trace decision-making.

Root cause: Git was available but not required, not taught, and not integrated into workflow. Students defaulted to familiar but inadequate tools.

Ephemeral storage platforms: Textile track demonstrated complete data loss due to dependence on commercial creative tools (Canva, Miro) tied to student accounts. When accounts expired, projects vanished.

Root cause: No institutional policy requiring durable storage formats or archival procedures. Students optimized for immediate convenience rather than long-term preservation.

Tool proliferation without integration: Students used disconnected tools (Word for writing, Excel for data, PowerPoint for presentations, email for sharing) creating fragmented workflows with manual transcription between systems—each transcription introducing potential errors.

Root cause: Lack of integrated reproducible research platform guiding students toward coherent toolchain.

Inadequate technical support: When students encountered technical problems (Git merge conflicts, Jupyter kernel crashes, dependency errors), resolution required specialized knowledge unavailable within teams.

Root cause: VCH Lab IT support focused on general infrastructure (network, accounts) rather than research-specific tools. No dedicated research computing support role.

6.3.2 6.3.2 Organizational Barriers: Missing Structure

Organizational barriers proved more significant than technical ones:

Educational model misalignment: The VCH Lab pedagogy emphasizes Theory U and design thinking—frameworks prioritizing observation, sensing, and creative emergence. Neither framework includes explicit reproducibility or documentation requirements.

As one student reflected: “We were never told documentation mattered for the grade, so we focused on making the final presentation look good.”

Root cause: Pedagogical philosophy values innovation outputs over research process transparency. Reproducibility framed as optional “best practice” rather than core competency.

Assessment criteria gaps: Analysis of GP1 assessment rubrics revealed that methodological transparency and verifiability are mentioned but constitute small fractions of total points. Students rationally allocate effort to criteria carrying most weight (problem definition, solution creativity, stakeholder impact).

Root cause: Assessment design reflects historical norms where applied research quality was judged by practitioner acceptance rather than scientific reproducibility.

Short project timelines: Four-month projects create pressure to “show results” quickly. Documentation is perceived as delaying tangible progress. As the CRM track demonstrated, teams prioritize concept development over process recording.

Root cause: Academic calendar constraints and desire to maximize student “value delivery” to industry partners.

High student turnover: Knowledge accumulated during one project cohort is lost when students graduate. Each cohort “reinvents” research practices. Institutional learning cannot accumulate.

Root cause: Structural characteristic of educational research labs—unavoidable but can be mitigated through persistent documentation infrastructure.

No quality assurance checkpoints: Projects lacked interim reviews focused on research quality. Supervisors reviewed “progress” (what results emerged) but not “process” (how results were derived). Problems only became visible at final submission.

Root cause: Limited faculty time for intensive project oversight; assumption that students possess research skills from prior coursework (often unfounded).

6.3.3 6.3.3 Contextual Barriers: Domain-Specific Challenges

Supply chain sustainability research presents unique challenges:

Stakeholder confidentiality: Industry partners often requested that specific company data, strategic plans, or competitive information not be publicly disclosed. While the five case studies did not exhibit confidentiality conflicts (due to their limited data collection), pilot project participants noted tension between transparency ideals and partner expectations.

Implication: Reproducible research frameworks must accommodate graduated transparency—methods and code can be open while preserving data confidentiality through synthetic data, aggregation, or access restrictions.

Methodological pluralism: Supply chain sustainability research integrates life cycle assessment, stakeholder interviews, systems modeling, regulatory analysis, and participatory design. No single reproducibility standard spans this methodological diversity.

Implication: Infrastructure must be flexible, supporting diverse methods rather than enforcing single approach. Templates provide starting points but cannot be prescriptive.

Contextual specificity: Supply chain interventions are deeply context-dependent. An effective practice in Dutch bakeries may not transfer to West African cacao supply chains. Reproducibility (generating same results in same context) differs from generalizability (transferring findings across contexts).

Implication: Reproducibility documentation must make context explicit—describing boundary conditions, assumptions, and scope limitations enabling appropriate transfer rather than naive replication.

Rapid evolution: Supply chains and regulatory environments change quickly. Research completed in 2023 references regulations amended by 2024. “Reproducibility” across time becomes challenging when external landscape shifts.

Implication: Version control and timestamp documentation create temporal reproducibility—ability to understand what was true at research time, even if no longer current.

6.3.4 6.3.4 Interactions Among Barrier Types

Barriers compound rather than simply adding:

Technical-organizational interaction: Version control infrastructure (technical) is only useful if organizational policies require its use and students receive training in its operation. VCH Lab’s JupyterHub (technical capability) was underutilized due to lack of training and integration into workflows (organizational gap).

Organizational-contextual interaction: Short timelines (organizational constraint) become more problematic when research methods are diverse and context-specific (contextual characteristics), requiring more rather than less documentation time.

Technical-contextual interaction: Tools optimized for computational reproducibility (Jupyter Notebooks) are less suited to qualitative methods common in sustainability research (contextual characteristic), requiring additional tool development (technical gap).

6.3.5 Answering RQ2: Systemic Barrier Architecture

Conclusion for RQ2:

Barriers to reproducibility form a **systemic architecture** rather than isolated obstacles:

Technical barriers (version control absence, ephemeral storage, tool fragmentation) are necessary but not sufficient explanations. They could be addressed through infrastructure investment.

Organizational barriers (misaligned pedagogy, inadequate assessment criteria, limited oversight) are deeper and more consequential. They shape whether technical capabilities are adopted.

Contextual barriers (confidentiality, methodological diversity, contextual specificity) are intrinsic to the domain and require adapted reproducibility frameworks rather than elimination.

Key insight: Barriers interact multiplicatively. Addressing technical barriers alone (providing Git repository) fails if organizational context does not require version control and students lack training. Addressing organizational barriers (changing assessment) fails if technical infrastructure is absent.

Implication for intervention design: Reproducibility solutions must be **holistic**, integrating technical infrastructure, organizational policy, educational training, and contextual adaptation simultaneously.

This finding validates (munaf2017)'s “manifesto for reproducible science” argument that cultural and systemic change is necessary—individual tools or training are insufficient.

6.4 Research Question 3: Confidentiality, Tool Limitations, and Transferability

RQ3: How do issues such as data confidentiality, tool limitations, and context specificity affect the transferability and credibility of reproducible research findings?

6.4.1 Data Confidentiality: Graduated Transparency

The research revealed that confidentiality need not eliminate reproducibility, but requires graduated transparency:

Levels of transparency:

1. **Full transparency:** Methods, code, data, and outputs all public (ideal for publicly-funded research with non-sensitive data)
2. **Methods transparency:** Research design, analytical procedures, and code public; data access restricted (suitable for business-sensitive data)
3. **Process transparency:** Internal documentation enables institutional reproduction even if external sharing prohibited (minimum for proprietary research)

Implementation in infrastructure:

The VCH Lab infrastructure implemented graduated transparency through:

- **Classification system:** Projects tagged as Public, Internal, or Confidential
- **Repository access control:** Public repositories for non-sensitive work, private repositories for confidential projects
- **Synthetic data substitution:** Templates for generating synthetic datasets matching statistical properties of confidential data, enabling external verification of analytical procedures even when real data cannot be shared
- **Embargo periods:** Temporary restrictions with scheduled public release

Findings from pilot project:

Industry partner initially concerned about “open research” accepted Internal classification with understanding that: - Methods and code would be open (demonstrating rigor) - Company-specific data would remain confidential - Aggregated findings publishable (contributing to knowledge) - Competitive insights protected

This graduated approach maintained reproducibility benefits (transparent methods, verifiable analysis) while addressing legitimate confidentiality concerns.

6.4.2 Tool Limitations: Workflow Integration Challenges

Tool limitations manifested in three areas:

Qualitative analysis gaps:

Jupyter Notebooks and RStudio are optimized for computational analysis. Qualitative research involving interview transcription, thematic coding, and interpretive synthesis requires different tools (NVivo, Atlas.ti, MaxQDA) with limited reproducibility affordances.

Attempted solution: Integration of qualitative data management templates (interview documentation, coding frameworks) into the infrastructure, alongside computational tools. This hybrid approach acknowledges that no single tool solves all reproducibility challenges.

Result: Partial success. Templates improved qualitative documentation but did not achieve computational notebook-level integration. Qualitative reproducibility remains an open challenge requiring further development.

Visual design constraints:

Supply chain projects often require visual communication (infographics, stakeholder maps, process diagrams). While tools like Canva and Miro excel at visual design, they lack reproducibility features (version control, scriptable generation, format durability).

Attempted solution: Encourage use of code-generated visualizations (Python matplotlib, R ggplot2, JavaScript D3.js) where possible. For necessary graphic design, require export to durable formats (SVG, PDF) with design files archived.

Result: Tension between design quality and reproducibility. Code-generated visualizations often less polished than manual designs. Students gravitate toward visual tools, requiring explicit intervention to preserve artifacts.

Collaboration friction:

Git's text-based diff and merge work well for code but poorly for notebooks (JSON format creates false conflicts) and not at all for binary files (Word documents, Excel spreadsheets).

Attempted solution: Tools like nbdime for notebook diffing, jupytertext for markdown conversion, and move toward plain-text workflows where possible.

Result: Improved but not eliminated friction. Merge conflicts remain confusing for novice users.

6.4.3 Context Specificity: Documentation of Boundary Conditions

The research found that context specificity does not prevent reproducibility but requires **explicit boundary condition documentation**:

What makes findings context-specific:

1. **Geographic factors:** Dutch regulatory environment, European supply chain structure, local industry norms
2. **Temporal factors:** Current state of technology, recent regulatory changes, market conditions
3. **Organizational factors:** Company size, maturity, culture, capabilities
4. **Methodological factors:** Available data, research access, stakeholder cooperation

Infrastructure approach:

The `project-metadata.yml` template requires explicit documentation of:

```
context:
  geographic_scope: "Netherlands, focus on Amsterdam region"
  temporal_scope: "Analysis conducted Feb-June 2024, data from 2023"
  organizational_context: "Mid-sized bakery chain (50-100 employees)"
  limitations:
    - "Results may not transfer to industrial-scale baking"
    - "Regulatory environment specific to EU"
    - "Farmer perspectives not directly captured"
```

Effect on transferability:

Explicit context documentation enables **informed transfer** rather than naive replication:

- Researchers in other contexts understand what might and might not transfer
- Comparative studies can systematically examine how context affects outcomes
- Meta-analyses can account for contextual variation

Effect on credibility:

Counter-intuitively, acknowledging context specificity **enhances** rather than diminishes credibility:

- Demonstrates researcher awareness of limitations
- Signals scientific humility rather than overconfidence
- Enables critical evaluation by readers
- Prevents false generalization

6.4.4 Trade-offs and Pragmatic Balancing

The research revealed inherent trade-offs requiring pragmatic balancing:

Trade-off 1: Transparency vs. Confidentiality

- Pure transparency maximizes verifiability but may be inappropriate or illegal for sensitive data
- Pure confidentiality protects interests but eliminates independent verification
- **Balance:** Graduated transparency with methods open, data appropriately protected

Trade-off 2: Tool sophistication vs. Accessibility

- Advanced reproducibility tools (Git, Docker, CI/CD) provide robust infrastructure but have steep learning curves
- Simple tools (shared drives, email) are accessible but lack reproducibility features
- **Balance:** Graduated tooling with templates and examples lowering barriers to advanced tools

Trade-off 3: Documentation completeness vs. Resource constraints

- Complete documentation of every decision and assumption would require unlimited time
- No documentation creates knowledge loss and irreproducibility
- **Balance:** Mandatory minimum documentation (templates, metadata) with optional deeper documentation for complex decisions

Trade-off 4: Standardization vs. Flexibility

- Rigid standards ensure consistency but may not fit diverse research methods
- Complete flexibility accommodates diversity but prevents comparison and integration
- **Balance:** Core standards (metadata, version control, file organization) with methodological flexibility within structure

6.4.5 Answering RQ3: Constructive Trade-offs

Conclusion for RQ3:

Data confidentiality, tool limitations, and context specificity **do not prevent reproducibility** but require **constructive trade-offs**:

Confidentiality → Graduated transparency: Methods and analytical procedures can be transparent even when data must be protected. Synthetic data, aggregation, access restrictions, and embargo periods enable appropriate transparency.

Tool limitations → **Hybrid infrastructure**: No single tool solves all reproducibility challenges. Infrastructure must integrate computational notebooks (quantitative), documentation templates (qualitative), version control (all), and appropriate exports from visual tools.

Context specificity → **Explicit boundary documentation**: Rather than seeking context-independent findings (often impossible in applied sustainability research), reproducibility requires making context explicit so others can understand transferability scope.

Impact on credibility: These accommodations to real-world constraints **enhance** rather than diminish credibility, by demonstrating: - Awareness of ethical responsibilities (confidentiality) - Methodological sophistication (appropriate tool selection) - Scientific humility (acknowledging limitations)

Key insight: “Perfect” reproducibility (full transparency, universal tools, context-free findings) is often neither possible nor desirable in applied research. “Appropriate” reproducibility (transparent within constraints, integrated tools, explicit context) is both achievable and valuable.

This extends (**freire2012?**)’s observation that reproducibility in business contexts requires adapted frameworks, not abandoned standards.

6.5 6.5 Cross-Cutting Themes and Insights

Beyond the specific research questions, several cross-cutting themes emerged:

6.5.1 6.5.1 The Primacy of Organizational Culture

Finding: Technical infrastructure is necessary but not sufficient for reproducibility. Organizational culture determines whether infrastructure is adopted.

Evidence: - VCH Lab had access to Git for years but no one used it until made mandatory - JupyterHub was underutilized until integrated into project templates - Pilot project succeeded largely due to coordinator commitment and assessment integration

Implication: Reproducibility initiatives must address culture change through: - Visible leadership commitment - Assessment alignment (grade what you value) - Success stories and champions - Community norms (“this is how we do research here”)

Connection to literature: Confirms (**nosek2015?**)’s argument that scientific culture reform is central to reproducibility progress.

6.5.2 6.5.2 The Documentation Paradox

Finding: Documentation is simultaneously the most important reproducibility practice and the least intrinsically rewarding.

Paradox components: - **Temporal asymmetry:** Documentation costs borne immediately (during busy project) but benefits accrue later (to future researchers, auditors, one's future self) - **Beneficiary asymmetry:** Documenter bears costs, but others (reproducers, reviewers) capture benefits - **Visibility asymmetry:** Undocumented research looks similar to documented research in final presentations—quality difference invisible to casual observers

Implication: Cannot rely on intrinsic motivation or voluntary compliance. Must create **structural incentives:** - Assessment explicitly rewards documentation quality - Templates make documentation easier (reduce friction) - Automation extracts documentation into visible outputs (progress dashboards, automated reports) - Peer recognition for well-documented projects

Connection to literature: Extends (stodden2018?)'s finding that reproducibility requires institutional incentive alignment.

6.5.3 6.5.3 The Expertise Distribution Problem

Finding: Reproducible research requires diverse expertise (research methods, software engineering, data management, domain knowledge) rarely concentrated in single individuals.

Challenge: VCH Lab students are developing business innovation expertise, not becoming research software engineers. Expecting students to master Git, Docker, Conda, Quarto, n8n, statistical methods, qualitative analysis, and domain knowledge within 4-month projects is unrealistic.

Implication: Infrastructure must **distribute expertise:** - Templates and automation embed software engineering best practices - Training provides “just enough” skills for specific tasks - Technical support handles complex troubleshooting - Documentation captures institutional knowledge across cohorts

Novel contribution: Applied research in educational contexts faces expertise distribution challenge not present in research institutes where teams include specialists. Infrastructure must compensate through embedded expertise.

6.5.4 6.5.4 The Learning Curve as Investment, Not Cost

Finding: Initial learning curve for reproducible research tools is steep, but returns compound over time.

Pilot project evidence: - Weeks 1-2: High frustration, low productivity, questioned value - Weeks 3-6: Increasing fluency, occasional “wow” moments when versioning saved work - Weeks 7-12: Natural workflow, cannot imagine working without version control

Student reflection: “At first I hated Git, but by the end I felt naked without it.”

Implication: Must communicate learning curve honestly and frame as **investment in professional skills**: - Version control, computational notebooks, documentation are marketable skills - Portfolio of reproducible projects demonstrates research quality to employers - Skills transfer to industry data science roles

Reframe: “This takes extra time” → “This builds career-relevant capabilities”

6.5.5 6.5.5 The Difference Between Reproducibility and Replicability

Clarification: This research focused on **reproducibility** (regenerating same results from same data/code) rather than **replicability** (conducting new study to validate findings).

Why this matters: - Reproducibility is *necessary* but *not sufficient* for scientific validity - Flawed analysis that is perfectly reproducible remains flawed - Reproducibility enables verification, which enables trust, which enables building upon findings

Implication: Reproducibility infrastructure must be complemented by methodological rigor training. Git version control does not substitute for sound research design.

Connection to literature: Aligns with (goodman2016?)’s distinction between methods reproducibility (this thesis focus) and results reproducibility (requires replication studies).

6.6 6.6 Theoretical Contributions

This research contributes to reproducibility theory in several ways:

6.6.1 Multi-Level Barrier Framework

The research proposes a **multi-level framework** for understanding reproducibility barriers:

Level 1: Individual (skills and motivation) - Technical skills (Git, Jupyter, statistical methods) - Methodological training (research design, documentation) - Intrinsic motivation (curiosity, scientific values)

Level 2: Project (resources and process) - Time available for documentation - Tool access and integration - Team collaboration dynamics - Quality assurance checkpoints

Level 3: Organizational (culture and incentives) - Pedagogical philosophy - Assessment criteria - Resource allocation - Leadership commitment - Community norms

Level 4: Institutional (infrastructure and policy) - Technical infrastructure provision - Data governance policies - Training programs - Long-term preservation systems - Inter-project knowledge transfer

Level 5: Field (disciplinary norms and standards) - Methodological traditions - Publication expectations - Funding requirements - Professional socialization

Key insight: Reproducibility failures typically involve misalignment across multiple levels. Single-level interventions (e.g., providing Git repository without training, or training without assessment alignment) often fail.

Theoretical contribution: Extends (munaf02017?)’s reproducibility framework by explicitly modeling level interactions and identifying minimum viable combinations for success.

6.6.2 Graduated Transparency Model

The research proposes **graduated transparency** as an alternative to binary open/closed dichotomy:

Traditional view: Research is either fully open (all materials public) or closed (nothing shared)

Graduated transparency view: Research transparency operates on multiple dimensions:

Dimension	Minimum	Moderate	Maximum
Methods	Research design described	Detailed procedures documented	Protocol pre-registered
Code	Pseudocode or description	Code shared with access restrictions	Code in public repository

Dimension	Minimum	Moderate	Maximum
Data	Data structure described	Synthetic or aggregated data shared	Full raw data public
Outputs	Results reported	Supplementary materials provided	Interactive reproductions published

Application context determines appropriate transparency level: - Publicly-funded academic research → maximum appropriate - Business-sensitive applied research → moderate with methods transparency prioritized - Proprietary commercial research → minimum with internal reproducibility maintained

Theoretical contribution: Resolves apparent tension between reproducibility ideals and applied research realities. Reproducibility is not abandoned in confidential contexts but adapted appropriately.

6.6.3 6.6.3 Infrastructure-as-Pedagogy Concept

The research demonstrates that **infrastructure can teach** through embedded design:

Traditional view: Infrastructure (tools, servers) is neutral platform; pedagogy happens separately through instruction

Infrastructure-as-pedagogy view: Design choices in infrastructure teach practices: - Git repository template with organized directory structure teaches file organization - Metadata templates with required fields teach documentation standards - Pre-commit hooks that check for large files teach data management - Quarto templates integrating code and narrative teach literate programming - n8n automation that validates documentation teaches quality assurance

Key insight: Students learn reproducibility practices by *using* well-designed infrastructure, not just by *hearing about* best practices.

Theoretical contribution: Extends “teaching in the environment” pedagogical theory (kearney2009?) into reproducible research domain. Infrastructure design becomes curriculum design.

6.6.4 6.6.4 Conditional Reproducibility Concept

The research introduces **conditional reproducibility**: reproducibility is not binary property but depends on context and resources:

Research is reproducible if and only if:

1. **Technical condition:** Computational environment can be reconstructed
2. **Methodological condition:** Procedures are documented sufficiently
3. **Data condition:** Data are available (or appropriately simulated)
4. **Skill condition:** Reproducer possesses necessary expertise
5. **Resource condition:** Reproducer has access to necessary infrastructure
6. **Temporal condition:** Reproduction attempted within dependencies' support lifetime

Implications: - “Is this research reproducible?” is incomplete question - Better question: “Under what conditions is this research reproducible?” - Reproducibility documentation should specify conditions explicitly

Theoretical contribution: Provides more nuanced reproducibility assessment framework than binary reproducible/irreproducible categorization.

6.7 6.7 Practical Implications

6.7.1 6.7.1 For VCH Lab and Windesheim University

Immediate actions (0-6 months):

1. **Policy integration:** Make reproducibility infrastructure use mandatory for all new VCH Lab projects
2. **Assessment alignment:** Revise GP1/GP2 rubrics to weight documentation and reproducibility more heavily
3. **Training integration:** Incorporate 2-hour reproducibility workshop into project onboarding
4. **Champion cultivation:** Identify early adopter students as peer mentors

Medium-term actions (6-18 months):

1. **Curriculum integration:** Embed reproducibility practices into earlier-year research methods courses
2. **Faculty development:** Train supervisors in reproducibility assessment
3. **Showcase development:** Create gallery of exemplar reproducible projects
4. **Industry communication:** Develop materials explaining reproducibility value to partners

Long-term actions (18+ months):

1. **Accreditation integration:** Use reproducibility capacity as quality indicator for program review
2. **Cross-university collaboration:** Share infrastructure with other Dutch universities of applied sciences

3. **Research output:** Publish methodological innovations from VCH Lab reproducibility initiative
4. **Continuous improvement:** Iterate infrastructure based on accumulated experience

6.7.2 6.7.2 For Applied Research Organizations

Lessons transfer to other practice-oriented research contexts:

Innovation labs in companies, NGOs, and government can adapt infrastructure by: - Implementing graduated transparency appropriate to confidentiality needs - Emphasizing process documentation for institutional learning - Creating templates for common project types - Building capacity through training and technical support

University applied research programs can adapt by: - Integrating reproducibility into program learning outcomes - Aligning assessment criteria with reproducibility standards - Investing in shared infrastructure (not expecting each project to build own) - Cultivating reproducibility culture through leadership commitment

Student-led research contexts can address unique challenges by: - Recognizing expertise limitations and embedding best practices in templates - Creating institutional memory through persistent documentation - Designing for knowledge transfer across cohorts - Treating reproducibility as core learning objective, not just research output quality

6.7.3 6.7.3 For Sustainability Research Community

Methodological pluralism: Reproducibility frameworks must accommodate diverse methods (LCA, stakeholder analysis, participatory design, systems modeling). No single approach works for all.

Contextual documentation: Supply chain sustainability research is inherently context-dependent. Reproducibility requires explicit boundary condition documentation, not context elimination.

Stakeholder engagement: Applied sustainability research involves practitioners, policymakers, and affected communities. Reproducibility documentation must be accessible to non-academic audiences.

Impact orientation: Sustainability research aims to change systems, not just describe them. Reproducibility enables verifying that interventions achieve intended impacts.

6.8 6.8 Methodological Reflections

6.8.1 6.8.1 Strengths of the Research Design

Case study depth: Systematic analysis of five diverse VCH Lab projects provided rich empirical foundation for understanding reproducibility failures

Design science integration: Combining diagnosis (case studies) with solution development (infrastructure) enabled both understanding and action

Reproducibility audit: Independent verification demonstrated infrastructure effectiveness, not just theoretical potential

Stakeholder involvement: Engagement with students, supervisors, coordinators, and industry partners ensured practical relevance

6.8.2 6.8.2 Limitations and Constraints

Single-institution focus: Findings specific to VCH Lab context; transferability to other settings requires careful assessment

Limited sample size: Five retrospective case studies and one pilot project; larger sample would strengthen generalizability

Short-term evaluation: Pilot project spanned 12 weeks; long-term sustainability and adoption remain uncertain

Researcher positionality: Developer of infrastructure evaluating that infrastructure creates potential bias (mitigated through independent auditor and transparent methods)

Technical scope: Focus on computational reproducibility; qualitative research reproducibility requires further development

6.8.3 6.8.3 Validity Considerations

Internal validity: Case study patterns consistent across five diverse projects; pilot project demonstrated causal link between infrastructure and reproducibility

External validity: Thick description and explicit scope conditions enable transferability assessment; findings likely applicable to similar applied research contexts

Construct validity: Multiple data sources (documents, interviews, audits) and explicit operationalization of reproducibility dimensions strengthen construct validity

Reliability: Detailed methodology documentation and preserved case study database enable replication

6.9 Chapter Summary

This discussion chapter addressed the three research questions:

RQ1: Computational notebooks *can* enable reproducibility in supply chain sustainability research, but enablement is **conditional** on appropriate research methods, institutional infrastructure, educational scaffolding, organizational incentives, and adequate timelines.

RQ2: Barriers to reproducibility are **systemic and interactive**, involving technical gaps (version control, storage), organizational misalignment (pedagogy, assessment, oversight), and contextual challenges (confidentiality, methodological diversity, context specificity). Addressing barriers requires holistic intervention across all dimensions simultaneously.

RQ3: Confidentiality, tool limitations, and context specificity do not prevent reproducibility but require **constructive trade-offs**: graduated transparency, hybrid tool infrastructures, and explicit boundary documentation. These accommodations enhance rather than diminish research credibility.

Cross-cutting insights: - Organizational culture trumps technical capability - Documentation requires structural incentives - Expertise must be distributed through infrastructure - Learning curves are investments, not costs - Reproducibility differs from replicability

Theoretical contributions: - Multi-level barrier framework - Graduated transparency model - Infrastructure-as-pedagogy concept - Conditional reproducibility framework

Practical implications provided for VCH Lab, applied research organizations, and sustainability research community.

The following chapter presents conclusions and actionable recommendations.

7 Chapter 7: Conclusions and Recommendations

7.1 7.1 Research Summary

This thesis investigated reproducible research challenges in applied supply chain sustainability innovation contexts, using the Value Chain Hackers Lab at Windesheim University as an empirical case. Through systematic analysis of five completed research tracks and development of a reproducible research infrastructure prototype, the research addressed three interconnected questions:

1. To what extent can computational notebooks enable reproducible research in real-world supply chain sustainability projects?
2. What are the main barriers to reproducibility in this domain?
3. How do confidentiality, tool limitations, and context specificity affect reproducibility?

Principal findings:

Finding 1: Reproducibility failures in VCH Lab are systemic, not individual All five analyzed research tracks exhibited serious reproducibility failures, ranging from partial documentation gaps (Cacao track: 1.5/3.0) to complete data volatility (Textile track: 0.0/3.0). Failures stemmed from absent institutional structures rather than student negligence.

Finding 2: Computational notebooks are necessary but not sufficient Jupyter Notebooks and RStudio provide powerful reproducibility capabilities, but effectiveness depends on institutional infrastructure, educational scaffolding, organizational incentives, and appropriate research contexts. Tools alone do not create reproducibility.

Finding 3: Barriers are multi-level and interactive Technical barriers (lacking version control), organizational barriers (misaligned pedagogy and assessment), and contextual barriers (methodological diversity, confidentiality needs) interact multiplicatively. Single-level interventions typically fail; holistic approaches are required.

Finding 4: Graduated transparency resolves confidentiality tensions Business-sensitive applied research can maintain methodological reproducibility even when full data sharing is inappropriate. Transparent methods, protected data, and explicit documentation of constraints balance openness and confidentiality.

Finding 5: Infrastructure can teach Well-designed infrastructure embeds best practices through templates, automation, and workflow integration. Students learn reproducibility by using infrastructure, not just through abstract instruction.

Finding 6: Reproducibility enhances applied research credibility Counter to concerns that reproducibility requirements would burden applied research, the pilot project demonstrated that reproducible practices improve stakeholder confidence, enable knowledge accumulation, and strengthen research impact.

7.2 7.2 Conclusions

7.2.1 7.2.1 Answering the Central Research Problem

The thesis addressed the overarching problem: **How can applied innovation labs build reproducible research capacity in supply chain sustainability contexts?**

Answer: Through **integrated socio-technical systems** combining:

1. **Technical infrastructure** (version control, computational notebooks, workflow automation)
2. **Organizational policy** (mandatory standards, assessment alignment, quality checkpoints)
3. **Educational programs** (training, templates, mentorship)
4. **Cultural change** (leadership commitment, community norms, visible benefits)
5. **Contextual adaptation** (graduated transparency, methodological flexibility, boundary documentation)

No single component suffices; all five must work together.

7.2.2 7.2.2 Theoretical Significance

This research contributes to reproducibility science by:

Extending reproducibility frameworks to applied contexts: Most reproducibility literature focuses on academic science. This thesis demonstrates how reproducibility principles must be adapted (not abandoned) for business innovation research with confidentiality constraints, stakeholder complexity, and practitioner audiences.

Demonstrating importance of organizational culture: Technical capabilities (tools, infrastructure) are necessary but insufficient. Organizational culture determines whether capabilities are adopted. This validates and extends organizational change theories into reproducible research domain.

Introducing conditional reproducibility concept: Reproducibility is not binary property but depends on multiple conditions (computational environment, documentation quality, data availability, reproducer skills, infrastructure access). This framework enables more nuanced reproducibility assessment.

Proposing infrastructure-as-pedagogy: Thoughtfully designed infrastructure teaches practices through use. This concept bridges technology design and educational theory.

7.2.3 7.2.3 Practical Significance

For Windesheim and VCH Lab: The infrastructure provides immediate pathway to improve research quality, enabling knowledge accumulation across project cohorts, strengthening credibility with industry partners, and supporting accreditation through demonstrable research rigor.

For applied research community: The methodology, infrastructure design, and lessons learned provide transferable guidance for other innovation labs, practice-oriented research programs, and educational research contexts facing similar reproducibility challenges.

For sustainability practitioners: Reproducible research practices enable verification of sustainability claims, reducing risk of implementing ineffective interventions and supporting evidence-based decision-making in supply chain sustainability.

7.2.4 7.2.4 Methodological Significance

The research demonstrates value of **combining retrospective case study analysis with prospective design science**. Case studies diagnose problems and identify requirements; design science develops and tests solutions. This combination generates both theoretical insights and actionable artifacts.

7.3 7.3 Recommendations

7.3.1 7.3.1 Recommendations for VCH Lab (Priority 1)

Recommendation 1.1: Mandate reproducibility infrastructure for all projects

Rationale: Voluntary adoption has failed repeatedly. Mandatory infrastructure use creates common baseline.

Implementation: - All new VCH Lab projects must initialize from provided Git repository template - Project registration includes infrastructure setup verification - Supervisors trained to assess reproducibility in addition to outputs

Recommendation 1.2: Align assessment criteria with reproducibility

Rationale: Students rationally allocate effort to graded dimensions. Documentation must carry substantial weight.

Implementation: - GP1 rubric: Add “Reproducibility and Documentation Quality” as 20% of grade - GP2 rubric: Require data and code availability for passing grade - GP3/4/5 rubrics: Include reproducibility audit in assessment

Recommendation 1.3: Integrate reproducibility training into onboarding

Rationale: Infrastructure without training creates frustration and abandonment.

Implementation: - Develop 2-hour workshop covering Git basics, Jupyter workflow, template usage - Create video tutorials for asynchronous learning - Assign peer mentors (prior cohort students) for troubleshooting - Embed training in first project week, not optional pre-work

Recommendation 1.4: Appoint reproducibility coordinator role

Rationale: Infrastructure requires ongoing maintenance, troubleshooting, and improvement.

Implementation: - Assign 0.2 FTE staff position for reproducibility coordination - Responsibilities: Technical support, template updates, training delivery, quality monitoring - Collaborate with IT support for server administration

Recommendation 1.5: Create reproducibility showcase

Rationale: Visible examples motivate adoption and provide learning resources.

Implementation: - Publish 3-5 exemplar projects publicly on VCH Lab website - Feature student interviews describing benefits - Use in marketing to prospective students and industry partners - Update annually with best projects

7.3.2 7.3.2 Recommendations for Windesheim University (Priority 2)

Recommendation 2.1: Embed reproducibility in curriculum

Rationale: Reproducibility skills should develop progressively, not only in capstone projects.

Implementation: - Year 1: Introduce version control and documentation basics - Year 2: Practice computational notebooks in research methods course - Year 3: Apply reproducibility in work placement - Year 4: Demonstrate mastery in graduation project

Recommendation 2.2: Develop institutional repository integration

Rationale: Long-term preservation requires connection to library infrastructure.

Implementation: - Configure automatic archiving from VCH Lab GitHub to Windesheim repository - Assign DOIs to completed projects via DataverseNL - Ensure metadata compatibility with national research infrastructure - Train library staff in reproducibility assessment

Recommendation 2.3: Share infrastructure across programs

Rationale: VCH Lab infrastructure benefits other applied research programs at Windesheim.

Implementation: - Package infrastructure as reusable template for other labs/programs - Provide training for faculty in other departments - Create community of practice around reproducible research - Leverage economies of scale in technical support

Recommendation 2.4: Include reproducibility in research integrity policy

Rationale: Formal policy establishes institutional commitment and expectations.

Implementation: - Draft reproducibility standards document - Integrate with existing research integrity and data management policies - Require reproducibility plans for all funded research projects - Include in faculty performance expectations

Recommendation 2.5: Invest in technical support capacity

Rationale: Sustainable infrastructure requires dedicated technical expertise.

Implementation: - Hire or train research software engineer (0.5-1.0 FTE) - Responsibilities span multiple research labs/programs - Provide professional development in reproducibility tools and practices - Establish support ticketing system for researcher assistance

7.3.3 7.3.3 Recommendations for Applied Research Community (Priority 3)

Recommendation 3.1: Adopt graduated transparency model

Rationale: Binary open/closed framework inappropriate for business-sensitive research.

Implementation: - Articulate transparency policies appropriate to organizational context - Prioritize methods transparency even when data must be protected - Use synthetic data, aggregation, and access controls strategically - Document confidentiality constraints explicitly

Recommendation 3.2: Develop domain-specific reproducibility standards

Rationale: Generic standards insufficient for methodologically diverse fields like sustainability research.

Implementation: - Convene stakeholders (researchers, practitioners, funders) to develop standards - Address qualitative methods, participatory research, and mixed-methods designs - Create discipline-specific templates and examples - Publish standards through professional associations

Recommendation 3.3: Recognize reproducibility in evaluation and promotion

Rationale: Researchers respond to career incentives.

Implementation: - Include reproducibility as criterion in hiring and promotion decisions - Value data/code publication alongside traditional publications - Award prizes for exemplary reproducibility practices - Feature reproducible research in institutional communications

Recommendation 3.4: Fund reproducibility infrastructure

Rationale: Infrastructure development requires resources often unavailable in project budgets.

Implementation: - Research funders allocate percentage of grants to reproducibility infrastructure - Support development of shared tools and platforms - Fund training programs and community building - Require reproducibility plans in grant applications

Recommendation 3.5: Collaborate across institutions

Rationale: Reproducibility challenges are common; solutions can be shared.

Implementation: - Dutch universities of applied sciences consortium on reproducibility - Share infrastructure, templates, training materials - Coordinate standards to enable cross-institutional collaboration - Jointly advocate for policy support

7.3.4 7.3.4 Recommendations for Sustainability Research Community (Priority 4)

Recommendation 4.1: Develop supply chain reproducibility guidelines

Rationale: Supply chain sustainability research has unique reproducibility requirements.

Implementation: - Working group to draft guidelines addressing confidentiality, context specificity, stakeholder engagement - Pilot guidelines across diverse supply chain contexts - Publish in open access journal - Promote adoption through professional networks

Recommendation 4.2: Create shared synthetic data resources

Rationale: Confidentiality often prevents data sharing; synthetic data enables methods verification.

Implementation: - Develop synthetic datasets representing common supply chain structures - Ensure statistical properties match real data distributions - Provide with documentation and use examples - Host on public repository

Recommendation 4.3: Establish reproducibility badges for sustainability research

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Rationale: Visible recognition motivates adoption.

Implementation: - Define badge criteria (Open Data, Open Code, Reproducible, Replicated) - Partner with sustainability journals to display badges - Create review guidelines for badge assessment - Track badge adoption over time

Recommendation 4.4: Integrate reproducibility into sustainability reporting standards

Rationale: Corporate sustainability claims should be verifiable.

Implementation: - Advocate for reproducibility requirements in frameworks like GRI, CDP, SASB - Develop guidance for making sustainability data FAIR - Provide tools for reproducible sustainability reporting - Recognize leaders through rankings and awards

7.3.5 7.3.5 Recommendations for Future Research (Priority 5)

Recommendation 5.1: Longitudinal study of VCH Lab reproducibility

Rationale: Short pilot project cannot assess long-term sustainability and impact.

Implementation: - Track reproducibility practices across multiple project cohorts (3-5 years) - Measure outcomes: knowledge accumulation, research efficiency, stakeholder satisfaction - Document challenges and adaptations over time - Publish findings to inform other institutions

Recommendation 5.2: Comparative study across innovation labs

Rationale: Generalizability requires evidence beyond single institution.

Implementation: - Partner with 5-10 innovation labs at other universities - Implement reproducibility infrastructure with local adaptations - Compare challenges, solutions, and outcomes across contexts - Identify universal principles and context-specific requirements

Recommendation 5.3: Qualitative research reproducibility methods

Rationale: Current infrastructure addresses computational work; qualitative methods require further development.

Implementation: - Design science project focused on qualitative reproducibility - Develop protocols for interview transcription, coding transparency, interpretive rigor - Create integrated qualitative data analysis reproducibility tools - Test with VCH Lab projects involving stakeholder research

Recommendation 5.4: Stakeholder perspectives on reproducibility

Rationale: Industry partner views on reproducibility are underexplored.

Implementation: - Interview study with VCH Lab industry partners - Investigate: How do partners perceive reproducibility value? What concerns exist? How does reproducibility affect collaboration? - Use findings to refine stakeholder communication strategies - Publish to inform research-practice partnerships

Recommendation 5.5: Economic analysis of reproducibility

Rationale: Cost-benefit analysis would support investment decisions.

Implementation: - Measure time investment for reproducibility practices - Assess efficiency gains from knowledge reuse - Quantify risk reduction from verifiable research - Calculate ROI for reproducibility infrastructure - Provide evidence for resource allocation decisions

7.4 7.4 Limitations and Future Directions

7.4.1 7.4.1 Limitations of This Research

Scope limitations: - Single institution (VCH Lab at Windesheim); findings may not generalize to all contexts - Five case studies; larger sample would strengthen patterns - Short pilot (12 weeks); long-term sustainability unproven - Computational focus; qualitative methods require further work

Methodological limitations: - Researcher also infrastructure developer (potential bias) - Single reproducibility auditor (multiple auditors would strengthen validation) - Retrospective case analysis (real-time observation would provide richer data) - Limited industry partner perspectives (more stakeholder voices needed)

Contextual limitations: - Dutch university of applied sciences context; international transferability unclear - Supply chain sustainability domain; other domains may differ significantly - Student-led projects; professional research teams face different challenges - Specific tools (Jupyter, Quarto); other tool ecosystems may behave differently

7.4.2 7.4.2 Future Research Directions

Direction 1: Scaling and sustainability Longitudinal research tracking VCH Lab reproducibility infrastructure over 3-5 years, assessing adoption, adaptation, impact, and sustainability. Key questions: Do benefits compound? How does infrastructure evolve? What organizational changes occur?

Direction 2: Generalization Comparative studies across multiple innovation labs in different countries, disciplines, and organizational contexts. Key questions: What findings are universal vs. context-specific? What adaptations are needed for different settings?

Direction 3: Qualitative methods Design science work developing reproducibility infrastructure specifically for qualitative and mixed-methods research common in sustainability studies. Key questions: How can interview analysis be transparent and verifiable? What role can AI play in qualitative reproducibility?

Direction 4: Economic impacts Rigorous cost-benefit analysis quantifying reproducibility's economic value through time savings, risk reduction, and enhanced credibility. Key questions: What is ROI of reproducibility infrastructure? How to optimize investment allocation?

Direction 5: Stakeholder engagement Research investigating how reproducibility affects research-practice partnerships, stakeholder trust, and knowledge mobilization. Key questions: How do practitioners perceive reproducibility value? What trade-offs exist between reproducibility and stakeholder accessibility?

Direction 6: AI and reproducibility Exploration of how generative AI tools can support reproducibility (automated documentation, code review, quality checking) while introducing new challenges (AI-generated content transparency, model reproducibility). Key questions: How can AI enhance reproducibility? What new risks emerge?

7.5 7.5 Contributions to Knowledge and Practice

This thesis makes several distinct contributions:

7.5.1 7.5.1 Empirical Contributions

- **First systematic documentation** of reproducibility barriers in student-led, practice-oriented, supply chain sustainability research
- **Quantitative assessment** of reproducibility failures across diverse project types
- **Demonstrated feasibility** of reproducible research infrastructure in applied educational context through successful pilot and independent audit

7.5.2 7.5.2 Theoretical Contributions

- **Multi-level barrier framework** explaining how technical, organizational, and contextual factors interact to create reproducibility challenges
- **Graduated transparency model** resolving tensions between openness ideals and confidentiality realities in business research
- **Conditional reproducibility concept** providing nuanced assessment framework beyond binary categorization
- **Infrastructure-as-pedagogy** showing how system design can teach practices

7.5.3 7.5.3 Methodological Contributions

- **Integration of case study and design science** demonstrating value of combining retrospective diagnosis with prospective solution development
- **Reproducibility audit protocol** for validating infrastructure effectiveness
- **EDRM application** to research infrastructure design in educational context

7.5.4 7.5.4 Practical Contributions

- **Working infrastructure prototype** ready for immediate use in VCH Lab and adaptable to other contexts
- **Template library** providing starting points for reproducible research practices
- **Implementation playbook** documenting lessons learned and pragmatic guidance
- **Training materials** including workshop design, video tutorials, and documentation

7.5.5 7.5.5 Policy Contributions

- **Actionable recommendations** for institutions, funders, professional associations
- **Standards proposals** for reproducibility in applied sustainability research
- **Policy frameworks** integrating reproducibility into research integrity and data governance

7.6 7.6 Final Reflections

Reproducibility is not just a technical challenge—it is fundamentally about **trust, cumulative knowledge, and scientific responsibility**.

In academic science, reproducibility enables verification and building upon prior work. In applied business innovation research, reproducibility serves additional functions:

Trust: Industry partners investing in research deserve confidence that findings are reliable **Learning:** Educational institutions must enable knowledge to accumulate across student cohorts **Impact:** Sustainability interventions should be based on verifiable evidence, not hopeful claims **Accountability:** Public and private investments in applied research require demonstrable rigor

The reproducibility crisis threatens all these functions. But crisis also creates opportunity for transformation.

This thesis demonstrates that reproducibility in applied contexts is **achievable**. It requires investment, commitment, and systemic change—but it is neither impossible nor

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impractical. The pilot project succeeded. Students, initially resistant, became advocates. Industry partners recognized value. Independent verification confirmed reproducibility.

Most importantly: **reproducibility enhances rather than constrains creativity**. Clear documentation enables bolder experimentation (knowing work can be recreated if it fails). Version control enables exploration without fear of losing prior work. Transparent methods invite collaboration and improvement.

The path forward requires collective action: - **Institutions** must invest in infrastructure and align incentives - **Educators** must teach reproducibility as core competency - **Students** must embrace documentation as craft, not burden - **Industry partners** must value process transparency alongside outcomes - **Funders** must support reproducibility infrastructure - **Professional communities** must establish and enforce standards

Windesheim's VCH Lab has the opportunity to lead. The infrastructure exists. The pilot succeeded. The recommendations are actionable.

The question is not whether reproducible applied research is possible, but whether we have the commitment to make it standard practice.

This thesis provides the evidence, the tools, and the roadmap. The next chapter belongs to VCH Lab, to Windesheim, and to the broader community of applied innovation researchers.

Let us build research systems worthy of the sustainability challenges we seek to address.

8 Chapter 8: Reflection

8.1 8.1 Personal Learning Journey

This graduation project has been transformative in developing my capabilities as an innovation professional. Reflecting on the BI Meta-skills framework, I recognize substantial growth across all five domains:

8.1.1 8.1.1 DEFINE: Uncovering the Right Problem

What I learned: At the start, I viewed reproducibility as primarily a technical problem—“students need Git training.” Through systematic case study analysis, I discovered the problem was far deeper: systemic organizational, cultural, and incentive misalignment.

The process of **properly defining** the reproducibility challenge required: - Resisting premature solution-jumping - Investigating root causes, not just symptoms - Considering multiple perspectives (students, supervisors, partners) - Recognizing that the “problem” was multi-layered

How I grew: I became more comfortable with ambiguity and complexity. Early in the project, I wanted clear, simple answers. By mid-project, I understood that valuable problems are rarely simple. The capacity to sit with complexity without forcing premature clarity is a meta-skill I will carry forward.

Evidence: - Case study analysis evolved through three iterations as understanding deepened - Research questions were refined twice based on emerging insights - Final problem definition (Chapter 1.4) is substantially more sophisticated than initial conception

8.1.2 8.1.2 DESIGN: Developing Solutions

What I learned: Solution design requires balancing multiple competing constraints: - Technical feasibility vs. user accessibility - Comprehensive functionality vs. manageable complexity - Ideal practices vs. practical realities - Standardization vs. flexibility

The iterative design process taught me that **first solutions are rarely optimal solutions**. The infrastructure went through four major iterations based on pilot feedback.

How I grew: I developed capacity for **design thinking under constraints**—not just imagining ideal solutions, but designing within real limitations (student expertise, supervisor time, server resources, organizational culture).

I also learned to design **for adoption**, not just for functionality. The infrastructure needed to work technically *and* be acceptable to users—these are distinct challenges requiring different design thinking.

Evidence: - Infrastructure architecture diagram (Chapter 5.2.2) reflects integration of multiple requirements - Template iterations based on user feedback - Graduated transparency model balances openness and confidentiality

8.1.3 8.1.3 EXECUTE: Implementation and Delivery

What I learned: Execution requires different skills than design. Technical implementation, stakeholder management, timeline management, quality assurance—each demands distinct competencies.

I learned the importance of **incremental delivery**. Rather than waiting for “perfect” infrastructure, I deployed minimum viable versions, gathered feedback, and iterated. This de-risked the project and kept stakeholders engaged.

How I grew: I developed **project management discipline**: maintaining task lists, tracking progress, managing dependencies, communicating transparently about challenges.

I also grew in **resilience**. Multiple setbacks occurred (pilot students initially resistant, reproducibility audit found bugs, timeline pressure intense). Maintaining momentum despite obstacles required emotional regulation and problem-solving persistence.

Evidence: - Pilot project successfully completed within 12-week timeline - Reproducibility audit demonstrated working infrastructure - Documentation completeness enabling future users - This thesis delivered on time despite substantial scope

8.1.4 8.1.4 LEARN: Monitoring and Adapting

What I learned: Research is inherently iterative. Every finding raises new questions. Every implementation reveals new challenges. The capacity to **learn from experience** and adapt accordingly is perhaps the most valuable meta-skill.

I learned to **embrace failure as information**. When templates proved too complex, that wasn’t “failure”—it was valuable feedback guiding simpler designs. When students struggled with Git, that wasn’t “incompetence”—it was signal that training needed improvement.

How I grew: I developed **reflective practice**: deliberately pausing to ask “what am I learning?” and “what should I do differently?” This reflection discipline transformed experiences into growth.

I also developed **comfort with uncertainty**. Research doesn’t provide clean answers. Many questions remain open. Being okay with “I don’t fully know yet” is maturity.

Evidence: - Multiple infrastructure iterations based on learnings - Explicit documentation of limitations and future directions (Chapter 7.4) - This reflection chapter demonstrating metacognitive awareness

8.1.5 8.1.5 LEAD: Managing Process and Stakeholders

What I learned: Leadership in research context means: - **Managing expectations:** Communicating realistic timelines and trade-offs - **Building buy-in:** Helping stakeholders understand value and address concerns - **Facilitating collaboration:** Coordinating across students, supervisors, IT support - **Championing vision:** Maintaining focus on reproducibility mission amid competing demands

How I grew: I learned to **lead through influence** rather than authority. I had no formal power over VCH Lab students or coordinators, yet needed their cooperation. This required persuasion, relationship-building, and demonstrating value.

I also developed **stakeholder empathy**. Initial frustration when students resisted infrastructure transformed into understanding their perspective: they’re learning business innovation, not becoming software engineers. This empathy led to better design choices.

Evidence: - Successful pilot project coordination across multiple stakeholders - Positive feedback from pilot participants despite initial resistance - Infrastructure adoption by VCH Lab coordinator for future projects - Recommendations (Chapter 7.3) demonstrating strategic thinking

8.2 8.2 Connection to Professional Ambitions

This graduation project aligns with and advances my career aspirations in several ways:

Short-term (0-2 years): Research software engineer or data scientist role

This project developed technical skills directly applicable: - Git/GitHub expertise for collaborative development - Python ecosystem (Jupyter, Pandas, etc.) for data analysis - CI/CD and automation (n8n, pre-commit hooks) - Documentation and communication (Quarto, technical writing) - Software project management

The reproducible research portfolio I created demonstrates capabilities to potential employers.

Medium-term (2-5 years): Research infrastructure specialist in academic or corporate R&D

This project provided foundation for specialization in research infrastructure: - Understanding researcher needs and pain points - Designing systems balancing technical requirements and user accessibility - Experience with institutional change and adoption challenges - Knowledge of reproducibility standards and best practices

I could pursue roles helping organizations build research capacity.

Long-term (5+ years): Consultant or entrepreneur in research infrastructure / open science

This thesis could be foundation for: - Consulting practice helping organizations implement reproducible research - Startup developing tools/services for applied research reproducibility - Academic career researching and teaching reproducible research practices - Leadership role in open science movement

8.3 8.3 Feedback Integration and Growth

Throughout the project, I received extensive feedback from multiple sources. Reflecting on how feedback shaped the work:

8.3.1 8.3.1 From VCH Lab Coordinator (Christiaan Verhoef)

Feedback: “The infrastructure is good technically, but you need to think about adoption barriers—culture matters more than tools.”

Integration: Added Chapter 6.5.1 on primacy of organizational culture. Revised recommendations to emphasize policy and incentive alignment alongside technical infrastructure. Developed “infrastructure-as-pedagogy” concept recognizing that tools must teach, not just function.

Growth: Learned that socio-technical systems require equal attention to social and technical dimensions. My initial engineer mindset underweighted cultural factors.

8.3.2 8.3.2 From Pilot Project Students

Feedback: “Git is confusing at first, but the templates helped a lot. We need more visual guides, not just text documentation.”

Integration: Created visual Git workflow diagrams. Developed video tutorials. Simplified README files with more screenshots and less technical jargon.

Growth: Learned that expertise curse makes me underestimate learner struggles. What seems “obvious” to me is foreign to novices. User empathy requires conscious effort.

8.3.3 8.3.3 From Reproducibility Auditor

Feedback: “Reproduction was successful, but some domain terminology was unclear to me as an outsider. More inline explanations would help interdisciplinary audiences.”

Integration: Added guidance in templates encouraging domain term definitions. Recommended “write for intelligent outsider” principle in documentation standards.

Growth: Recognized tension between efficient communication (experts understand shorthand) and reproducibility (outsiders need explanation). Leaned toward reproducibility priority.

8.3.4 8.3.4 From Thesis Supervisor

Feedback on early drafts: “You need stronger theoretical framing. Connect your empirical findings to reproducibility literature more explicitly.”

Integration: Expanded literature review (Chapter 2). Added theoretical contributions section (Chapter 6.6). Explicitly cited supporting literature throughout discussion.

Growth: Developed academic writing discipline: making theory-practice connections explicit, situating work within scholarly conversations, distinguishing contribution from confirmation.

8.3.5 8.3.5 From GP1 Resit Feedback

Original GP1 feedback: “The analysis at micro, meso and macro level is still limited. The near future situation becomes a bit more clear in paragraph 1.3, but not the urgency. What happens if this project would not be executed?”

Integration: Chapter 1 (Introduction) now includes: - Comprehensive micro, meso, macro analysis (sections 1.2-1.4) - Explicit urgency argument with consequences of inaction (section 1.3) - Near-future scenarios (addressed in Chapter 6 discussion)

Growth: Learned to think systematically across levels of analysis and temporally (past-present-future). Initial tendency to focus on immediate present expanded to contextual thinking.

8.4 8.4 Challenges and How They Were Overcome

8.4.1 Challenge 1: Scope Management

Problem: Initial project scope was enormous—could have written thesis on any of: reproducibility theory, infrastructure design, case study analysis, qualitative methods, stakeholder perspectives, economic analysis.

Solution: Ruthlessly prioritized. Focused on computational reproducibility in VCH Lab context. Explicitly scoped out qualitative methods and economic analysis as future work.

Learning: Completeness is impossible; strategic incompleteness is necessary. Better to do focused work deeply than superficial work broadly.

8.4.2 Challenge 2: Technical Debugging

Problem: Multiple infrastructure components (JupyterHub, Git, Quarto, n8n) with complex interactions. When things broke, diagnosis was difficult.

Solution: Systematic debugging methodology: isolate components, test individually, reintegrate incrementally. Maintained detailed logs of configuration changes. Built test cases for reproducibility.

Learning: Complex systems require disciplined debugging practices. Intuition and trial-and-error are insufficient. Systematic approaches save time.

8.4.3 Challenge 3: Stakeholder Resistance

Problem: Initial pilot student reactions included skepticism: “This seems like extra work,” “Why can’t we just use Google Drive?”

Solution: Empathy and demonstration. Acknowledged learning curve honestly. Showed early benefits (recovered deleted work via Git, generated report from notebooks automatically). Positioned reproducibility as career-relevant skill, not academic obligation.

Learning: Change resistance is legitimate, not obstinate. Meet people where they are. Benefits must be visible early to sustain engagement.

8.4.4 Challenge 4: Time Pressure

Problem: Balancing deep research (requires time) with graduation deadline (fixed) created constant tension.

Solution: Aggressive timeline with milestone tracking. Prioritized “good enough” over “perfect” in early stages, with plan to iterate. Accepted that some questions would remain open.

Learning: Perfect is enemy of done. Strategic satisficing (sufficient + satisfying) is valuable skill. Knowing what to defer matters as much as knowing what to do.

8.4.5 Challenge 5: Researcher Bias

Problem: As infrastructure developer, I had vested interest in “proving” it worked, creating potential confirmation bias.

Solution: Engaged independent reproducibility auditor. Actively sought disconfirming evidence in case studies. Documented limitations explicitly. Triangulated findings across data sources.

Learning: Reflexivity (examining own biases) is essential research practice. Transparency about positionality and limitations enhances rather than diminishes credibility.

8.5 Ethical Reflections

Several ethical dimensions merit reflection:

8.5.1 Student Work Analysis

Analyzing prior students’ work (case studies) raised ethical questions: Am I being fair? Am I respecting their efforts? Am I causing potential embarrassment?

Resolution: - Anonymized student identities - Framed failures as systemic, not individual (“absent infrastructure” not “lazy students”) - Acknowledged students operated within constraints not of their making - Emphasized learning purpose, not judgment

Reflection: Critique and respect can coexist. Systems critique is different from personal criticism.

8.5.2 8.5.2 Reproducibility Requirements

Imposing reproducibility requirements on students raises fairness questions: Is this adding burden? Are we changing expectations mid-program?

Resolution: - Phased implementation (new cohorts, not mid-project changes) - Provided substantial support (training, templates, troubleshooting) - Positioned as skill development, not compliance - Ensured assessment alignment so effort rewarded

Reflection: New standards must be introduced equitably with adequate support.

8.5.3 8.5.3 Open Science and Confidentiality

Tension between open science ideals (full transparency) and business realities (confidentiality) required ethical navigation.

Resolution: - Graduated transparency model respecting legitimate confidentiality - Prioritized methods transparency even when data must be protected - Explicit documentation of constraints - Rejected false dichotomy (either fully open or not reproducible)

Reflection: Ethical research means navigating competing values thoughtfully, not applying absolute principles rigidly.

8.6 8.6 What I Would Do Differently

With hindsight, several aspects could have been improved:

Start infrastructure development earlier: I conducted case studies first, then built infrastructure. Parallel development would have saved time and enabled earlier pilot testing.

Engage more users in pilot: Single pilot project (3 students) provided valuable feedback but limited sample size. Multiple concurrent pilots would have accelerated learning.

Record detailed decision rationale throughout: While I documented decisions, deeper real-time rationale capture would have enriched thesis discussion. I reconstructed some reasoning retrospectively.

Involve industry partners more directly: Limited direct engagement with business stakeholders. Their perspectives on reproducibility value would have strengthened the research.

Develop qualitative methods further: Computational focus left qualitative reproducibility underdeveloped. Allocating more time here would have produced more complete solution.

8.7 8.7 Advice for Future Graduation Students

Based on this experience, I would advise future students:

Define before designing: Invest time understanding the problem deeply. Premature solutions are rarely optimal. The temptation to “start building” is strong—resist it initially.

Embrace iteration: First drafts are terrible. First prototypes break. First research questions are often wrong. That’s normal. Build iteration into timeline.

Seek feedback early and often: Don’t wait until work is “ready” to show others. Rough drafts generate better feedback than polished but misdirected work.

Manage your energy: Research marathons, not sprints. Sustainable pace matters more than heroic bursts followed by burnout. Schedule rest.

Document as you go: Future-you will forget today’s reasoning. Write it down now. Research logs, decision notes, reflection journals—all valuable.

Connect with your “why”: When motivation wanes (it will), reconnect with why the work matters. My “why” was enabling better sustainability research—that sustained me through difficult moments.

Balance depth and breadth: You cannot research everything. Focus depth where it matters most. Acknowledge but don’t fully explore every tangent.

Enjoy the process: Graduation project is rare opportunity for deep, self-directed investigation. Despite stress, I genuinely enjoyed the intellectual challenge. Try to savor it.

8.8 8.8 Future Directions for My Development

Looking forward, areas for continued growth:

Technical depth: While I developed solid foundation in reproducibility infrastructure, deepening expertise in specific areas (Docker/Kubernetes, CI/CD platforms, advanced Git workflows) would enhance capability.

Qualitative methods: Deliberately building qualitative research skills would complement quantitative strengths and enable more holistic research practice.

Change management: Infrastructure success depends on organizational adoption. Studying change management, organizational behavior, and innovation diffusion would strengthen intervention design.

Communication skills: Technical concepts must be communicated to diverse audiences. Continued development of writing, presenting, and visual communication is life-long endeavor.

Domain knowledge: Deeper understanding of supply chain management, sustainability science, and organizational theory would enable richer contributions to applied research.

8.9 8.9 Gratitude and Acknowledgments

This project succeeded through support of many people:

Christiaan Verhoef, VCH Lab coordinator at Windesheim University of Applied Sciences, provided technical mentorship, infrastructure access, stakeholder connections, and continuous encouragement. His vision for research quality in applied contexts inspired this work.

My thesis supervisor at Inholland University of Applied Sciences challenged my thinking, demanded rigor, and helped situate work within academic context.

Pilot project students, who trusted new infrastructure, provided candid feedback, and demonstrated reproducibility's value through their success.

Reproducibility auditor, who invested time verifying infrastructure and provided honest assessment.

VCH Lab faculty and staff, who welcomed external researcher, shared materials, and engaged with findings.

Family and friends, who tolerated months of “I can’t, I have to work on my thesis” with patience and support.

To all of you: thank you. This thesis exists because of your contributions.

8.10 8.10 Closing Reflection

This graduation project challenged me intellectually, technically, and personally. I am not the same researcher, professional, or person who began this journey.

I understand now that innovation—genuine innovation—is socio-technical, not merely technical. The hardest problems involve people, culture, incentives, and context. Tools are easy; change is hard.

I recognize that valuable research requires humility: acknowledging what you don’t know, seeking diverse perspectives, accepting criticism, revising assumptions. Certainty is comfortable but rarely justified.

8 Chapter 8: Reflection

I appreciate that impact requires patience and persistence. Reproducibility infrastructure will not transform VCH Lab overnight. Cultural change is slow. But incremental progress accumulates. The seeds planted here may bear fruit over years.

Most importantly, I learned that work is meaningful when it serves something larger than oneself. This thesis contributes—modestly but genuinely—to better research practices that enable more effective sustainability interventions. That matters.

As I transition from student to professional, I carry these lessons forward: define carefully, design thoughtfully, execute persistently, learn continuously, and lead with both competence and humility.

The reproducibility crisis is real. The challenges are substantial. But solutions exist. With commitment, collaboration, and courage, we can build research systems worthy of the urgent sustainability challenges humanity faces.

This thesis is one small contribution to that essential work.

9 References

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10.1 Appendix A: Case Study Documentation Audit Scores

Table A.1: Detailed Documentation Audit Results

Case	Methodological Transparency	Data Integrity	Verifiability	Archival Sustainability	Overall Score
Bakery	0/3	0/3	0/3	1/3	0.25/3.0
CRM	1/3	1/3	0/3	1/3	0.75/3.0
Orga	1/3	N/A	1/3	2/3	1.33/3.0
Textile	0/3	0/3	0/3	0/3	0.00/3.0
Cacao	1/3	2/3	1/3	2/3	1.50/3.0
Pilot	3/3	3/3	3/3	3/3	3.00/3.0

10.2 Appendix B: Infrastructure Repository Structure

Complete directory structure for VCH Lab reproducible research project template:

```
vch-project-template/
  .git/                    # Version control database
  .github/
    workflows/
      validate.yml        # CI/CD validation workflow
      publish.yml         # Quarto publish workflow
  .gitignore              # Git ignore patterns
  .pre-commit-config.yaml # Pre-commit hooks configuration
  README.md               # Project overview and quick start
  LICENSE                 # CC-BY-4.0 for docs, MIT for code
  CITATION.cff            # Citation metadata
  CONTRIBUTING.md         # Contribution guidelines
  CODE_OF_CONDUCT.md      # Community standards
  environment.yml          # Conda environment
  requirements.txt         # Python dependencies
  renv.lock               # R dependencies (if applicable)
```

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```
data/
  raw/                                # Immutable raw data
    README.md                         # Data provenance documentation
    .gitkeep
  processed/                          # Cleaned, transformed data
    README.md
    .gitkeep
  external/                          # Third-party data
    README.md
    .gitkeep
  synthetic/                         # Synthetic data for public sharing
    README.md
    .gitkeep
notebooks/
  00-template.ipynb                  # Template with best practices
  01-data-collection.ipynb
  02-data-preprocessing.ipynb
  03-exploratory-analysis.ipynb
  04-modeling.ipynb
  05-validation.ipynb
  README.md
scripts/
  utils.py                          # Reusable functions
  data_processing.py
  analysis.py
  visualization.py
docs/
  index.qmd                         # Project homepage
  project-plan.qmd                  # GP1 deliverable
  research-report.qmd               # GP2 deliverable
  final-report.qmd                  # GP3/4/5 deliverables
  appendices.qmd
  _quarto.yml                       # Quarto configuration
  references.bib                    # Bibliography
  apa.csl                           # Citation style
  assets/
    images/
    templates/
outputs/
  figures/
    README.md
  tables/
    README.md
```

```

reports/
  README.md
metadata/
  project-metadata.yml          # Dublin Core metadata
  data-dictionary.csv          # Variable documentation
  ethics/
    consent-template.md
    README.md
  provenance/
    README.md
tests/
  test_data_processing.py
  test_analysis.py
.vch/
  templates/                   # VCH Lab specific templates
  workflows/                   # n8n workflow exports
  config.yml                   # Project configuration

```

10.3 Appendix C: Interview Protocol Example

Semi-Structured Interview Guide: VCH Lab Coordinator

Estimated duration: 60-90 minutes

Opening (5 minutes) - Introduce research purpose and process - Confirm consent and recording permission - Explain how data will be used - Invite questions

Section 1: Current VCH Lab Research Practices (20 minutes)

1. Can you describe how a typical VCH Lab project progresses from initiation to completion?
 - *Probe:* What are the key milestones?
 - *Probe:* Who is involved at each stage?
2. What guidance do students receive about research methodology and documentation?
 - *Probe:* Are there required templates or standards?
 - *Probe:* How is quality assessed?
3. What tools and platforms do students typically use?
 - *Probe:* Are certain tools required vs. optional?
 - *Probe:* What technical support is available?
4. How are completed projects archived and made accessible for future reference?

Section 2: Reproducibility Challenges (25 minutes)

5. In your experience, what challenges do students face in documenting their research process?
 - *Probe:* Time constraints?
 - *Probe:* Technical skills?
 - *Probe:* Motivation/incentives?
6. Can you share examples of past projects where documentation was particularly strong or weak?
 - *Probe:* What made the difference?
7. How do industry partners react to the research outputs from VCH Lab?
 - *Probe:* Do they ever question methodology or request more transparency?
8. What happens when a new project wants to build on prior work?
 - *Probe:* How easy is it to access and understand previous projects?

Section 3: Infrastructure and Support (15 minutes)

9. What technical infrastructure is currently available to support reproducible research?
 - *Probe:* Compute resources, storage, software
10. What barriers exist to implementing better reproducibility practices?
 - *Probe:* Technical, organizational, cultural, resource
11. What support would be most valuable for improving research quality?

Section 4: Future Vision (15 minutes)

12. If you could design the ideal reproducible research system for VCH Lab, what would it look like?
13. What concerns do you have about implementing more structured reproducibility requirements?
 - *Probe:* Student resistance, supervisor workload, industry partner expectations
14. How would you measure success of reproducibility infrastructure?

Closing (5 minutes) - Invite any additional thoughts or questions - Explain next steps in the research - Offer to share findings - Thank participant

Post-Interview Notes Template - Setting and atmosphere - Non-verbal observations - Topics that generated strong reactions - Potential follow-up questions - Initial reflections

10.4 Appendix D: Reproducibility Audit Checklist

VCH Lab Reproducibility Audit Protocol

Auditor Name: _____ *Project:* _____ -
Date: _____

Phase 1: Environment Setup (Target: 30 minutes)

Repository cloned successfully README instructions clear and complete Dependencies documented Environment installed without errors Test data accessible

Time taken: _____ minutes *Issues encountered:*

Phase 2: Code Execution (Target: 45 minutes)

Notebooks run top-to-bottom without modification No undocumented dependencies Execution completes without errors Runtime warnings documented/explained

Time taken: _____ minutes *Issues encountered:*

Phase 3: Results Verification (Target: 30 minutes)

Numerical results match documentation ($< 1\%$ difference) Visualizations substantially equivalent Tables contain expected data Summary statistics consistent

Discrepancies noted:

Phase 4: Comprehension Assessment (Target: 45 minutes)

Research question clear Methodology understandable Analytical steps logical Conclusions supported by results Limitations acknowledged

Comprehension challenges:

Phase 5: Documentation Evaluation (Target: 30 minutes)

Code well-commented Data documentation complete Metadata present and accurate Dependencies specified Version information included

Documentation gaps:

Overall Assessment

Reproducibility Score: Fully reproducible (all phases successful, < 3 hours) Mostly reproducible (minor issues, 3-5 hours) Partially reproducible (significant issues, 5-8 hours) Not reproducible (unable to complete, > 8 hours)

Recommendations for improvement:

- 1.
- 2.
- 3.

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Strengths:

- 1.
- 2.
- 3.

Auditor Signature: _____ Date: _____

End of Thesis

Total approximate word count: ~50,000 words across all chapters Estimated page count: 60-65 pages at standard academic formatting